Adjoint-Faithful Fourier Neural Operator for a Baroclinic Double Gyre

A minimum-data summer internship project plan

Core idea: first establish a frozen, adapted Bire-architecture baseline on the same MITgcm trajectories used by AF–FNO. Then improve the operator with modern forward losses and a small set of forward perturbation–response pairs. Generate true MITgcm adjoints only after the emulators are frozen, and use them as a blind test.

Revised technical proposal — 29 July 2026

Ten-week scope; official MITgcm baroclinic-gyre tutorial; NeuralOperator 2.0 implementation

Contents

1 Recommendation in one page

Recommended path. Build an **adjoint-faithful FNO (AF–FNO)** on the official MITgcm baroclinic double-gyre tutorial at 1°. Before changing the architecture or loss, train a frozen **A0 adapted Bire FNO** on exactly the same trajectories. This tests transfer of the paper's qualitative trends—useful short forecasts, stable rollouts, and sensitivity to wind amplitude—without claiming a bitwise reproduction of its undocumented 0.25° dataset. Then introduce the modern A–D ladder: A and B are controlled baselines, C is selected for forward fidelity, and D adds forward-only response supervision. Do not put a single MITgcm adjoint in training or model selection.

The minimum credible programme is:

- **Three long production regimes**: low, control, and high wind amplitude.
- **Two short held-out wind runs**: the intermediate amplitudes used by Bire et al.; inference-only and not 20 years long.
- **One frozen architecture-transfer baseline (A0)**: the recovered Bire FNO core, adapted only where the 62×62, 46-state benchmark forces a change. It is trained and evaluated before the modern A–D models.
- **224 ten-day perturbed restarts**: 128 training, 32 validation, and 64 inference. They provide local tangent information from forward MITgcm only and total about 6.1 model-years.
- **Twelve true-adjoint runs**: three smooth objectives, two horizons, and two unseen initial states, opened only after the model and criteria are frozen.

This separates four questions: Does the Bire architecture transfer to the documented tutorial data? Does modernization improve the forward emulator? Did response supervision improve the local input–output map? Does automatic differentiation through it yield a useful adjoint? The defensible claim is deliberately limited: *adjoint fidelity over specified horizons, objectives, states, and a tested control subspace.*

No-adjoint-training contract. MITgcm adjoints are excluded from training, hyperparameter tuning, early stopping, loss selection, perturbation-amplitude selection, and architecture selection. The test is run only after checksums of the frozen model and evaluation script are recorded.

2 Research question and hypotheses

Let the MITgcm time-Δt map be

$$x_{n+1} = \mathcal{M}_{\Delta t}(x_n, u_n),$$

where x is the state and u contains forcing and controls. The FNO $\mathcal{F}_\theta$ approximates the same map. For a scalar objective $J(x_N)$,

$$\lambda_n^{\text{FNO}} = \left(\frac{\partial \mathcal{F}_\theta}{\partial x_n}\right)^T \lambda_{n+1}^{\text{FNO}}, \qquad \lambda_N = \nabla_{x_N} J.$$

Primary question. Can local derivative information obtained exclusively from short forward perturbation experiments make these gradients agree with MITgcm adjoints more closely than a state-accurate forward FNO?

Hypotheses: (1) the adapted A0 baseline reproduces the qualitative Bire forecast and wind-response trends on the coarser tutorial; (2) a forward-selected modern FNO improves forecast, rollout, boundary, and resolved-scale behavior relative to A0; (3) state accuracy alone does not ensure adjoint accuracy; and (4) adding a local response loss to a forward-credible model improves gradient direction without adjoint labels or material loss of forward skill.

3 Why local perturbation–response information is the right approach

3.1 A small state error can hide a wrong derivative

Suppose $\mathcal{M}(x) = x$, but an emulator learns

$$\mathcal{F}(x) = x + \delta \sin(kx), \qquad \mathcal{F}'(x) = 1 + \delta k \cos(kx).$$

The state error never exceeds δ, yet the derivative error can be large when k is large. A value-only loss controls the height of a learned curve at sampled points; it does not adequately control its slope. An adjoint is built from those slopes.

> **Intuitive ocean example**
> Two emulators predict sea-surface height ten days ahead to within one centimetre. In one, a small extra wind-stress patch strengthens the western boundary current; in the other, it weakens it. Their forecast scores can be almost identical, but a gradient-based optimization steps in opposite directions. One perturbed forward run reveals the physical response sign without computing an adjoint.

3.2 Forward responses teach Jacobian actions

At an anchor $z = (x, u)$ and normalized direction v, run the model from z and $z + \varepsilon v$:

$$r_M(z; v, \varepsilon) = \frac{\mathcal{M}_{\Delta t}(z + \varepsilon v) - \mathcal{M}_{\Delta t}(z)}{\varepsilon} \approx D\mathcal{M}_{\Delta t}(z)v.$$

Match it with the emulator secant

$$r_\theta(z; v, \varepsilon) = \frac{\mathcal{F}_\theta(z + \varepsilon v) - \mathcal{F}_\theta(z)}{\varepsilon}.$$

This constrains a Jacobian–vector product (JVP) using only forward simulations and ordinary first-order training backpropagation. The link to adjoints is duality:

$$\underbrace{v^T D\mathcal{M}(z)^T q}_{\text{gradient projected on } v} \quad = \quad \underbrace{q^T D\mathcal{M}(z) v}_{\text{objective response along } v} \quad .$$

Thus, matching $D\mathcal{M}v$ for relevant directions also matches projections of adjoints for many later objectives q. The full Jacobian remains too large to identify; the claim must stay within the sampled smooth wind and state subspace.

4 Benchmark choice: trend, not hidden dataset

The public `oceanfourcast` repository is a conceptual starting point, not an exact reproduction package. Its training defaults differ from the paper, its README points to external data, and its loader predicts ten selected channels including diagnostic pressure and streamfunction. Reimplement the method in the maintained NeuralOperator library rather than preserve old dependencies [4, 5].

Table 1: Published setup, official tutorial, and proposed benchmark.

	Bire et al.	**Official tutorial**	**Proposed**
Grid	0.25°, 248×248	1°, 60×60 ocean cells plus land border	Tutorial first; fine grid only as stretch
Vertical	15 levels, about 2 km	15 levels, 1.8 km	Tutorial levels
Time step	300 s	1200 s	Tutorial value
τ_0 ($\mathrm{N\,m^{-2}}$)	0.075, 0.0875, 0.100, 0.1125, 0.125	Default 0.100	Same five amplitudes
Training winds	Low, control, high	Not applicable	Low, control, high
Held-out winds	Two intermediates	Not applicable	Same two; short runs
Goal	Forward emulation	Numerical tutorial	Forward and local-adjoint fidelity

This removes uncertainty about an unavailable initialization and pipeline, reduces the horizontal cell count by about 16, and retains the relevant baroclinic physics. The tutorial notes that the deep ocean is still adjusting after 100 years; stationarity must therefore be measured for the variables and horizons used [3].

4.1 A0: a time-boxed Bire-architecture transfer baseline

Do not make the incomplete public repository a prerequisite for the project. Audit the paper, repository, and recovered implementation once; record unresolved discrepancies; then freeze A0 and label it *adapted* rather than exact. Implement it through the public NeuralOperator 2.0 `SpectralConv` interface while retaining the recovered paper structure: channel-diagonal Fourier multipliers, explicit pointwise channel mixing, three Fourier blocks, width 128, alternating sine/cosine position fields, and paper-style one-step state loss and optimizer schedule.

All models must use the same S0–S2 source states, chronological splits, masks, and training-only normalization. Only adaptations forced by the documented benchmark are allowed in A0:

Run a 20–100-sample overfit test, one ten-day development training, and the locked low/control/high evaluation. Do not tune A0 after opening held-out I1/I2 results. If the recovered baseline cannot learn the small sample after the declared learning-rate fallbacks, report that result and proceed; repairing historical code must not delay the AF–FNO experiment. Separate five- and 30-day A0 models are optional follow-ups only when needed to compare a specific Bire lead-time panel.

5 Minimum MITgcm simulation programme

Table 2: Declared adaptations from the Bire configuration to A0.

Element	**Bire / recovered setting**	**A0 on the tutorial**
Horizontal grid	248×248 at $0.25°$	62×62 including the land rim
Dynamic state	Ten selected prognostic/diagnostic outputs	All 46 Markov-state channels; wind is the additional input
Fourier modes	$[64, 64]$	$[16, 16]$, preserving the approximate mode-to-grid ratio
Coarse pretraining	Stride 8, yielding 31×31	Not used in the frozen A0; this is a recorded departure, not an inferred reproduction
Forecast target	Normalized next state	Retain for A0; Models A–D predict a normalized residual
Software	Historical `oceanfourcast` environment	Recovered architecture on pinned NeuralOperator 2.0

Table 3: Minimum trajectories. Durations are planning budgets; diagnostics decide whether equilibration is adequate.

ID	τ_0	**Duration**	**Initialization**	**Use**	**Output**
S0	0.100	100 y spin-up; final 10 y production	Tutorial initial state	Control data; branch checkpoint	Annual restart; daily production
S1	0.075	5 y adjust + 10 y production	S0 checkpoint	Low-wind data	Daily production
S2	0.125	5 y adjust + 10 y production	S0 checkpoint	High-wind data	Daily production
E0	0.100	10 y extension	S0 year-110 pickup	Version-2 control coverage	Daily production
E1	0.075	10 y extension	S1 local year-15 pickup	Version-2 low-wind coverage	Daily production
E2	0.125	10 y extension	S2 local year-15 pickup	Version-2 high-wind coverage	Daily production
I1	0.0875	1 y	Unseen S0-derived state	Interpolation inference only	Daily
I2	0.1125	1 y	Different unseen state	Interpolation inference only	Daily
Total		**172 model-years**	Original 142 plus 30 evidence-authorized extension years		

5.1 Five original trajectory jobs plus three authorized extensions

After year 50, inspect five-year trends of heat content, surface kinetic energy, basin-integrated transport, and level-mean temperature. Stop at year 100 only if the upper-ocean quantities are statistically stationary; otherwise extend the run or narrow the first claim. The 1° grid and four-times-larger time step should be much cheaper per model-year than Bire's setup, but wall-clock/model-year must be benchmarked.

5.2 Chronological data splits

Use years 1–7 for training, year 8 for validation, and years 9–10 for inference within each ten-year S0–S2 production segment. Insert 90-day gaps at boundaries.

Table 4: Data roles. Pair counts are approximate for daily output and a ten-day target.

Dataset	**Source**	**Approx. size**	**Permitted use**
Trajectory training	S0–S2 years 1–7	7,500 pairs	Fit weights
Trajectory validation	S0–S2 year 8	1,000 pairs	Tune and early-stop
Known-wind inference	S0–S2 years 9–10	2,100 pairs	Final scores only
Unseen-wind inference	I1 and I2	700 pairs	Final interpolation only
Response training	16 anchors × 8 directions	128 endpoints	Fit response loss
Response validation	4 anchors × 8 directions	32 endpoints	Tune loss and ε
Response inference	4 anchors × 8 directions × 2 signs	64 endpoints	Blind central differences
True adjoints	3 costs × 2 horizons × 2 states	12 integrations	Blind comparison only

The 224 perturbed restarts add 2,240 model-days, about 6.1 model-years. Save only initial/final states, configuration hash, and perturbation definition. A 46-channel 62×62 float32 snapshot is about 0.7 MB; 30 daily production years are of order 8 GB before compression.

5.3 Realised splits, and Bire's own arrangement

The plan above predates the evidence and is retained as the original design. Three arrangements have since been run on the pooled S0–S2 regimes, and they differ enough that any reported skill must name the one it was measured under.

Table 5: Split arrangements actually run, on 9,000-day trajectory-v3 regimes.

Arrangement	**Train**	**Validation**	**Inference**	**Notes**
Stored (interleaved)	0–2519, 3690–6209	interior	6660–7199	Training surrounds validation in time; supports an interpolation claim only
Chronological	0–5039	5130–6389	6480–8999	180-day buffers; evaluation 1,441–1,961 days past training
Bire §3.2	0–5999	6000–7199	6200–7199	No buffers; inference nested in validation; evaluation 264–980 days past training

Bire's arrangement is quoted directly: *"Final 6000 timesteps . . . for training. The next 1200 . . . for validation. The final 1000 timesteps are for inference (testing set),"* with *"We do not use a third held-out test data set."* Since $6000 + 1200 = 7200$ is the whole record, the inference block is the last 1,000 days of validation, not a third partition, and there are no buffers. Leakage is prevented structurally: a three-step training rollout qualifies only if t, $t + 10$ and $t + 20$ all lie inside training, so the latest start is 5,969 and its last target lands on 5,999.

Two consequences should be carried forward. First, the arrangement is the dominant control on measured skill: with architecture, objective, optimizer, seed and dataset held fixed, short-lead error against persistence moves from 0.686/0.659/0.623 under the chronological split to 0.435/0.378/0.296 under Bire's. The operative variable is how far the evaluation block sits beyond the end of training, measured against this project's 248-day S0 SSH e-folding scale. Second, a nested inference block gives a weaker independence guarantee than a disjoint one, so results under Bire's protocol replicate the paper's setting rather than establishing prospective skill.

Model visibility and evaluation truth are tracked separately. Under Bire's protocol the model sees strictly the 7,200-day record, enforced in code on training targets, validation rollouts and inference starts; days 7200–8999 carry no split code and are used only as lead-matched truth, which is what reproduces the day-2,000 ground-truth column of the paper's Figure 7. Bire obtain the same separation across simulations, holding two of five entirely out; we obtain it along time.

5.4 Eight directions per response anchor

Table 6: Minimum smooth perturbation basis.

Count	Direction family	Purpose
1	Prescribed cosine-wind amplitude	Directly tests the Bire control axis
1	Smooth meridional wind phase/shape mode	Separates amplitude from pattern learning
2	Smooth localized wind envelopes	Probe western/southern and northern/interior control
4	Balanced, low-wavenumber state directions	Training-only EOFs or filtered anomalies; rotate among anchors

Pilot standardized amplitudes 10^{-3}, 3×10^{-3}, and 10^{-2}. Select the largest for which a central-difference linearity check is within 5–10%, then freeze the rule by direction family. Use randomly signed one-sided perturbations for training and central differences for the blind response test.

6 State, forcing, and preprocessing

6.1 Predict a Markov state, not independent diagnostics

Place the prognostic variables on a documented common tracer grid and stack vertical levels as channels:

$$x = [U_{1:15}, V_{1:15}, \Theta_{1:15}, \eta], \qquad C_{\text{out}} = 46.$$

Vertical velocity, pressure, and streamfunction are not independent prognostic channels. Streamfunction is diagnosed from all 15 U levels. For the present z-coordinate, linear-EOS, constant-salinity configuration, pressure is reconstructed as the exact MITgcm diagnostic `PHIHYD`, i.e. hydrostatic pressure anomaly divided by ρ_0, in $\mathrm{m^2\,s^{-2}}$:

$$\frac{\rho'_k}{\rho_0} = -\alpha_T(\Theta_k - T_{\mathrm{ref},k}), \qquad \Phi'_k = g\eta + \mathcal{H}_k\left(\frac{\rho'}{\rho_0}\right).$$

Here $\mathcal{H}$ is the fixed finite-difference vertical integration implemented in `calc_phi_hyd.F`; the $g\eta$ term matches `diags_phi_hyd.F`. Report Φ' at Bire's zero-based surface, mid-depth, and bottom indices $0, 7, 14$. Physical pressure anomaly in Pa is $\rho_0\Phi'$. This map is linear, differentiable, and has a fixed transpose $\mathcal{H}^T$ for later adjoint objectives. It therefore adds evaluation views but no new learned information or output channels. If salinity later becomes active and nonconstant, add its 15 levels and update the equation of state rather than silently omitting it.

The implementation was checked against the archived S0 MITgcm final-state `PH` dump at iteration 2,851,200 over all 15 levels and 3,600 wet cells. Maximum absolute error was $3.89 \times 10^{-4}\ \mathrm{m^2\,s^{-2}}$ and RMSE was $9.75 \times 10^{-6}\ \mathrm{m^2\,s^{-2}}$, passing the predeclared 10^{-3} maximum-error tolerance. The reproducible report is `outputs/af_fno/pressure_validation_v1/pressure_validation.json`.

The barotropic circulation diagnostic has a different verification contract. From raw C-grid faces,

$$U_{\mathrm{bt}} = \sum_k U_k \Delta z_k, \qquad V_{\mathrm{bt}} = \sum_k V_k \Delta z_k, \qquad \Psi_U = -\int U_{\mathrm{bt}}\, dy, \qquad \Psi_V = \int V_{\mathrm{bt}}\, dx.$$

The production output is explicitly the *U-derived barotropic transport streamfunction*, matching the tutorial construction. A training-only audit uses one fixed 360-day year from each S0–S2 regime and compares independent raw-face U and V prefix integrals over daily and 5/30/90/360-day means. Version 1 verified the project centered-U operator exactly but also exposed that centering U and V paths onto north and east faces creates a spurious 10.9% comparison. Corrected version-2 contract SHA-256 `88ff2f20fe1a...dea170` retained the fixed year and thresholds and collocated both paths at identical C-grid corner indices. All regimes pass: daily p90 relative mismatch is at most 3.25×10^{-5}, daily mean absolute mismatch is 1.14–3.49×10^{-4} Sv, 360-day mismatch is 4.74–6.00×10^{-5} Sv, spatial correlation is effectively one, and all outer-boundary normal transports are zero. Artifacts are under `outputs/af_fno/streamfunction_consistency_v2/`.

This is strong numerical path consistency, not the same algebraic statement as the pressure reconstruction. With the linear free surface,

$$\partial_t \eta + \nabla_h \cdot \mathbf{U}_{\mathrm{bt}} = 0,$$

so an instantaneous daily field may retain a very small divergence when SSH changes. Use the U-derived label for instantaneous products, allow the validated transport streamfunction interpretation for daily and time-mean comparisons in this configuration, and retain this caveat in figures and gates.

Inputs comprise the 46 dynamic channels, wind stress, and normalized longitude, latitude, wet mask, and distance-to-wall fields. Record every C-grid interpolation operator. Compare gradients only after applying the exact transpose of those operators so the FNO and MITgcm sensitivities share a native control basis.

Compute channel means and scales from trajectory training only, mask land, and store the affine transform with the model. Gradient comparisons must undo normalization:

$$\widetilde{x}_i = (x_i - \mu_i)/\sigma_i \quad \Longrightarrow \quad \frac{\partial J}{\partial x_i} = \frac{1}{\sigma_i}\frac{\partial J}{\partial \widetilde{x}_i}.$$

Models A–D predict the normalized ten-day residual $\Delta\widetilde{x}_n = \widetilde{x}_{n+1} - \widetilde{x}_n$. This gives the network an explicit identity path and focuses capacity on change. A0 alone retains the recovered paper-style normalized next-state target; this intentional difference belongs to the architecture-transfer baseline and must be stated in every A0 comparison.

7 Modern adjoint-faithful FNO

7.1 Starting implementation

After freezing A0, use the maintained `neuraloperator` package and pin its release and commit. Begin the common A–D ladder with a modest dense FNO:

Table 7: Starting AF–FNO configuration. Validation uses only trajectories and forward responses.

Choice	**Starting value**	**Reason / limited search**
Forecast interval	10 days	Economical response runs; useful scale in Bire et al.
Fourier modes	$[16, 16]$	Compare 12, 16, 24 after inspecting spectra; respect Nyquist
Hidden channels	32	Increase to 64 only if a 20–100-sample overfit test fails
FNO layers	4	Current library default; compact enough for ablation
Local branch	`conv_bias_kernel=3`	Complements global mixing near boundaries
Boundary treatment	10% domain padding	Reduces periodic FFT wrap in a closed basin
Geometry	Grid embedding, mask, wall distance	Makes solid boundaries explicit
Channel MLP	Enabled	Mixes the vertical state channels locally
Precision	float32	Avoid mixed precision in the initial derivative study
Factorization	Dense FNO first	TFNO is a memory fallback, not an initial confounder

Select mode count from the resolved power spectrum: too few modes oversmooth the western boundary current; too many learn aliased fluctuations. Domain padding or Fourier continuation is relevant because the basin is not periodic. These choices follow the NeuralOperator 2.0 practical guide rather than the outdated oceanfourcast environment [2].

7.2 Baseline ladder and controlled losses

For a one-step pair and response direction, use

$$\mathcal{L} = \mathcal{L}_{\text{state}} + \lambda_{\text{roll}}\mathcal{L}_{\text{roll}} + \lambda_{\text{resp}}\mathcal{L}_{\text{resp}} + \lambda_{\text{spec}}\mathcal{L}_{\text{spec}} + \lambda_{\text{bnd}}\mathcal{L}_{\text{bnd}}, \tag{1}$$

$$\mathcal{L}_{\text{resp}} = \frac{\|W_r(r_\theta - r_M)\|_2^2}{\|W_r r_M\|_2^2 + \epsilon_0}. \tag{2}$$

$\mathcal{L}_{\text{state}}$ is masked relative L_2; $\mathcal{L}_{\text{roll}}$ is a three-step (30-day) unrolled loss; $\mathcal{L}_{\text{spec}}$ compares binned spectra; and $\mathcal{L}_{\text{bnd}}$ upweights wet cells near the western wall. An H^1-style spatial loss may supplement the spectral term, but it is not a substitute for $\mathcal{L}_{\text{resp}}$: spatial output derivatives and derivatives of the input–output operator are different objects.

Models A and B remain frozen controls; B is not retroactively retuned. Model C is the only forward-selection stage, and its search is limited and recorded below. Model D copies only a Model C configuration that has passed its complete forward gate; rejected validation-search v1 does not qualify. It then preserves that configuration's architecture, preprocessing, and forward objective and adds $\mathcal{L}_{\text{resp}}$. Increase λ_{resp} only on response validation data until response error improves without worsening C's validation state error by more than 5%. Lock all weights before viewing any true adjoint.

Table 8: Required architecture-transfer baseline and controlled model ladder.

Model	Training signal	Purpose	Expected result
A0	Adapted Bire architecture and paper-style one-step loss	Test architecture transfer before modernization	Qualitative Bire trend on the common data
A	Modern dense FNO + one-step residual state loss	Controlled modern baseline	Comparable state skill; derivatives may be poor
B	Model A + rollout + spectral/boundary losses	Isolate modern forward losses from response data	Better rollout and boundary state skill
C	Forward-selected dense FNO + balanced state/increment, rollout, tapered spectral, and boundary losses	Establish a forward-credible emulator before derivative training	Pass the complete forward gate
D	Frozen Model C design + local response loss	Test the AF–FNO hypothesis without asking derivative data to repair the forward model	Preserve C's state skill; improve responses and adjoints

7.3 Model C: predeclared forward-selection protocol

The preliminary A/B results show that architecture plausibility is not enough: temperature and SSH can be worse than persistence even when the 46-channel aggregate loss improves. Following the practical FNO workflow in Duruisseaux et al. [2], Model C therefore separates representation, optimization, and data adequacy rather than changing all three at once.

Model C implementation contract opened 23 July 2026. The first implementation retains the 51 declared raw inputs, NeuralOperator grid lifting, dense float32 spectral weights, Channel MLP, four-layer width-32 starting point, 10% padding, and parallel 3×3 local branch. It replaces B's channel-aggregated objective with equal U/V/temperature/SSH group terms, scales increment error with training-only per-channel ten-day RMS, and compares spectra of increments on the exact 60×60 wet rectangle after a Hann taper. Initial forward/backward checks contain 630,262 parameters at width 32 and 2,443,542 at width 64. Width 64 is therefore a conditional capacity test, not the assumed default. Training-only C1a calibration subsequently froze increment/rollout/spectral/boundary coefficients $0.001/0.15/10^{-5}/0.065$; these scales precede all validation and architecture selection.

C0 training-only audit, 23 July 2026

The sealed audit verified the dataset metadata hash and read only 7,530 training pairs. The per-channel normalized ten-day increment RMS spans a factor of 147, so channelwise increment scaling is necessary. At $(16, 16)$, the global Fourier path contains 8.8%, 24.0%, 67.8%, and 85.6% of tapered U, V, temperature, and SSH increment energy; full-state fractions all exceed 99.6%. The local branch is therefore essential, and more modes alone cannot represent most velocity-increment energy. The $(24, 16)$ candidate captures 12.5%/32.9% for U/V, compared with 9.0%/25.6% for $(16, 24)$; it was added before validation was opened. Increment autocorrelation implies only about 278–398 effective samples across the three regimes, with S1 much more correlated than S0/S2. This supports regime-balanced learning curves, but not immediate replacement of version-1 data.

C1a loss-gradient calibration, 23 July 2026

Complex-safe CPU job 285116 completed in 4m39s using only the same 96 balanced training rollouts. After its 20-epoch core warmup, state/increment/rollout/spectral/boundary gradient norms were 1.965/830.5/6.484/40,763/7.411. The automatically clipped proposal was rejected: coefficients 0.01 would still make increment and spectral gradients dominate. Unclipped targets were 0.001183/0.1515/1.205×10^{-5}/0.06629; the rounded frozen coefficients above produce weighted gradient ratios 0.423/0.495/0.207/0.245 relative to state. The very small spectral coefficient does not remove spectral supervision because its raw gradient is about 20,700 times the state gradient. Pending GPU replica 284858 was cancelled with zero runtime. No candidate checkpoint or validation metric was produced.

C1b–C1d memorization diagnosis, 23–26 July 2026

Training-only job 285192 completed both controlled 160-epoch attempts in 58m30s. At learning rate 0.001 it passed total, spectral, state-group, rollout, and boundary criteria, but temperature/SSH increment errors were 1.802/2.888 times persistence on the exact 96 records. Learning rate 0.0005 improved these to 1.285/2.032; U/V were already 0.254/0.442 times persistence. Therefore C1b is rejected specifically for slow-field increments despite a 97.7% total reduction, 99.1% spectral reduction, 0.0304 maximum state-group error, 0.0312 rollout, and 0.0397 boundary error.

The lower learning rate achieved its best total at the final epoch, so C1c job 285265 fixed the data, loss, architecture, seed, and batch order and extended 160 to 320 epochs. At its epoch-315 minimum, U/V/temperature/SSH ratios were 0.206/0.353/0.979/1.219; all other criteria and exact reload passed. Width-64 GPU job 285325 also completed 320 epochs but was rejected at 0.145/0.292/1.083/1.385 at its selected epoch, and its best balanced SSH ratio was 1.302. More duration and four times the parameter count therefore did not remove the SSH failure. Every evaluated checkpoint and both rejected diagnostic checkpoints are retained; no validation data were opened.

Late-objective audit and accepted optimization schedule, 26 July 2026

GPU array job 287581 independently rechecked all 64 evaluated epochs in each 320-epoch history and differentiated the full 96-record mean loss at both late checkpoints. Increment contributed only 5.7% and 7.2% as much value as state at widths 32 and 64, but its weighted gradient norm was already 0.422 and 0.406 times the state gradient. The increment gradient was dominated by SSH and aligned with state (cosines 0.966 and 0.927), while late SSH errors fluctuated strongly. This evidence supports an optimization diagnosis rather than another capacity increase. Operational CPU fallback job 287556 measured the same short width-64 workload in 436/272/253 seconds at 4/8/16 cores; 16 cores is the measured fallback, while GPU remains primary and backend replication is not a scientific priority.

Bounded array job 287583 then isolated two training-only changes. Raising only the increment coefficient from 0.001 to 0.0025 (loss v2) was rejected; its best balanced SSH ratio remained 1.328. Keeping loss v1 and reducing the learning rate once from 0.0005 to 0.0001 after epoch 240 passed the complete gate at epoch 315: U/V/temperature/SSH were 0.211/0.360/0.794/0.990 times persistence, total/spectral reductions were 98.24%/99.82%, maximum state-group error was 0.0248, rollout/boundary were 0.0231/0.0301, and three-step reload was bitwise exact. Only epoch 315 passed every group, so the result authorizes the bounded validation search but is not itself a validated or frozen final Model C.

Model C validation-search contract frozen before validation metrics, 26 July 2026. The machine-readable contract `config/model_c_validation_search_v1.json` has SHA-256 `9e1d44299ae6...c4c6f6`. It compares the five declared mode pairs at widths 32/64 and four layers/10% padding in four from-scratch successive-halving rounds: $10 \to 5 \to 3 \to 2 \to 1$. Chronology and optimizer steps increase together as 25%/1,920, 50%/3,840, 75%/5,760, and 100%/7,680. Every round decays learning rate 0.0005 to 0.0001 at 75% of its step budget; the full round exactly transfers the accepted 96-record schedule by optimizer step rather than by dataset-dependent epoch count.
Eight predeclared checkpoint fractions are scored on all 780 ten-day validation pairs and 16 fixed 180-day-capable starts per regime. Ranking is lexicographic: require all four physical-unit U/V/temperature/SSH RMSE ratios below persistence, minimize their worst ratio, then minimize the arithmetic mean of full-domain, western-boundary, and Hann-tapered local-increment spectral ratios at 30/90/180 days. Search seed 20260723 is fixed. The final architecture must then pass every ten-day group independently for seeds 20260723/20260724/20260725 with exact reload before any inference data open. Padding 20% or six layers may be tested only for the two finalists and only under the contract's explicit wall-leakage or training-underfit triggers; no combined variant is allowed. At contract freeze, only split metadata had been inspected—no validation state metric, inference, intermediate-wind, response, or adjoint datum had been read.

Completed bounded validation search and three-seed decision, 26 July 2026

GPU jobs 287604/288466/289379/289915 completed every candidate in the four from-scratch $10 \to 5 \to 3 \to 2 \to 1$ rounds. Stage-manifest SHA-256 values are `40b54530d7d9...12d2f05`, `a2d639f8940c...d5e382`, `9bbd4fc4ce3c...f30e17`, and `453c05ec408f...2fea2`. The selected full-resource architecture was $(24, 16)$, width 64, four layers, and 10% padding. Its U/V/temperature/SSH validation ratios were 0.1473/0.3547/0.6597/1.00014, with physics score 22.84; the other finalist, $(16, 24)$, width 64, gave 0.1577/0.3723/0.6353/1.02477 and 28.22. Training worst-group ratios 0.915/0.941 and boundary/full-domain ratios 0.667/0.639 left both predeclared conditional triggers false, so neither six layers nor 20% padding was tested.
Final-seed GPU array 290382 then retrained the selected architecture from scratch for seeds 20260723/20260724/20260725. All jobs completed with scheduler exit zero, matching source/data/normalizer contracts, sealed later-data flags, and bitwise-exact three-step reload. Every seed beat persistence for U, V, and temperature, but SSH ratios were 1.00014/1.00383/1.03818. The frozen rule is strict, so the immutable decision records `scientifically_rejected_three_seed_gate`, `configuration_frozen=false`, and `inference_authorized=false`. Decision SHA-256 is `e19d502442fc...72155`; internal freeze-contract SHA-256 is `a44622cee685...73628`. The full-domain mean group ratio also grows from 0.704–0.765 at 30 days to 3.21–3.45 at 90 days and 53.54–61.90 at 180 days. Therefore validation-search v1 is scientifically rejected, not rounded into a near-threshold pass, and no inference or response/adjoint archive opens.

The following list preserves the prospectively declared protocol that produced this decision; it is not authority to continue tuning after the v1 rejection.

1. **Training-only diagnostics.** Compute tapered two-dimensional spectra of normalized states and ten-day increments by variable group, verify padding and mode indexing, run a stronger 96-sample memorization test, and log both raw and weighted loss terms and their gradient norms. Report U, V, temperature, and SSH separately; the 15/15/15/1 channel multiplicity must not decide the fit.

2. **Repair the objective before enlarging the network.** Start from B's dense four-layer, width-32 architecture. Use an equal-weight mean of the four group-normalized state errors, add a group-balanced normalized-increment error, retain the three-step rollout, and compute spectral loss after wet-field extension and a fixed taper so the land discontinuity is not mistaken for unresolved physics. Normalize boundary and spectral terms by group and choose their weights from the training diagnostics, then freeze the weights for candidate comparison.

3. **Small representation search.** Compare mode pairs $(12, 12), (16, 16), (16, 24), (24, 16), (24, 24)$ and widths 32 and 64 with four layers and 10% padding. Use successive halving on fixed training/validation blocks. For only the two best candidates, test padding 20% and six layers if the failure diagnostic specifically indicates wall leakage or insufficient capacity. Dense float32 spectral weights, lifting, projection, Channel MLP, mask, wall distance, and the local 3×3 branch remain mandatory; factorization is a memory fallback.

4. **Freeze by validation, then test once.** Train the selected configuration with three fixed seeds. Candidate ranking is lexicographic: first require every U/V/temperature/SSH ten-day validation RMSE ratio to beat persistence; next minimize the worst group ratio; then minimize the predeclared mean of 30-, 90-, and 180-day rollout, boundary, and resolved-spectrum scores. Inference years, I1/I2, response inference, and adjoints remain sealed until the configuration, seeds, code, and normalizer hashes are recorded. The complete 10–360-day package is run only after that freeze.

This search is deliberately not "try until a test plot looks good." A candidate that reduces the aggregate validation loss but remains worse than persistence for any state group is not selected. Conversely, increasing Fourier modes is not automatic: the 62-point grid limits meaningful frequencies, and added modes are accepted only when training-only increment spectra and held validation metrics support them. For the real-valued NeuralOperator convolution, a declared `n_modes` $= (m_y, m_x)$ retains m_y centered Fourier rows but only $m_x/2+1$ nonnegative real-FFT columns internally. Every spectral audit and report uses that exact convention rather than interpreting 16×16 as 256 independent complex coefficients.

Calibration backend rule. CUDA is an efficiency choice, not a scientific requirement for the C1a loss-gradient calibration. Its 20 epochs on 96 training rollouts completed on eight CPU cores in 4m39s. The CPU and GPU replicas must use the same data, records, seed, architecture, and loss code but write separate artifacts. If both complete, compare raw terms, gradient ordering, and proposed weighted contributions; modest floating-point differences are expected, whereas an order-of-magnitude disagreement blocks the loss freeze and requires diagnosis. Global gradient norms are accumulated in float64 for real gradients and complex128 for complex spectral-weight gradients even though model arithmetic remains float32/complex64; this prevents diagnostic overflow without discarding the imaginary component. Neither backend output is a candidate checkpoint or validation result.

7.4 Data-adequacy and expansion gate

The existing version-1 trajectories are sufficient to diagnose and select Model C: across S0–S2 they provide 7,530 overlapping ten-day training pairs, 780 validation pairs, and 1,860 sealed inference pairs. They are not 7,530 independent realizations, because adjacent daily pairs overlap strongly. More consecutive years would therefore increase the raw count much faster than the effective sample size.

Before generating replacement data, run learning curves on chronological 25, 50, 75, and 100% subsets, estimate integral autocorrelation times for the four state groups, compare training and validation error by group, and measure the three-seed spread. The evidence is considered *data-limited* only when a candidate can fit the training blocks, validation skill improves consistently with more chronology, and between-seed or between-regime generalization remains the dominant error. Poor training fit or insensitivity to added chronology points instead to optimization, capacity, or the objective and does not authorize more MITgcm years.

> **Post-search data-adequacy decision, 27 July 2026**
>
> The selected $(24, 16)$, width-64 candidate fits the full training chronology (worst-group ratios 0.915–0.928 across the three final seeds), while its validation worst-group ratio improves monotonically with chronological coverage: 2.168, 1.483, 1.192, and 1.00014 at 25/50/75/100%. The three final SSH ratios span 1.00014–1.03818. Contract `config/model_c_data_adequacy_v1.json`, SHA-256 `e185301947b4...53c84a9`, was frozen before per-regime checkpoint metrics. GPU job 290415 passed all four rules. Temperature/SSH state-RMS effective sample totals are only 7.59/7.69; S0, S1, and S2 all have positive median validation–training SSH gaps. Aggregate SSH block-bootstrap intervals include persistence.
>
> The audit also prevents a data-only interpretation. Median S1 SSH ratios are 1.354 on training and 1.508 on validation, while S0/S2 are below persistence. Thus `trajectories_v2` is authorized, but the low-wind objective/capacity imbalance and long-rollout instability must still be addressed prospectively.

Expansion contract `config/trajectories_v2_expansion.json`, SHA-256 `7ee2ae10330a...fe045ed`, adds ten exact, continuous production years to each immutable S0–S2 endpoint in a new `mitgcm_v2/` root. Exact continuation is chosen over an artificial pickup perturbation so the added states remain on precisely the same forced attractors and the record length approaches the reference study. Raw coverage doubles, but effective coverage is recomputed after conversion and is not assumed to double automatically. Version 1 and every conclusion based on it remain immutable.

The 7,200-record-per-regime successor split contains two seven-year training blocks, the original sealed inference block, and fresh validation/inference blocks separated by 90-day buffers. The already-open v1 validation block is excluded from successor selection. This yields 15,060 training pairs, 780 fresh validation pairs, and 3,450 sealed inference pairs across S0–S2.

> **Trajectory-v2 completion and coverage gate, 27 July 2026**
>
> Array job 290439 completed the three exact continuations in 15m10s–15m31s. Each produced 3,600 daily `dynState/surfState` records, the endpoint pickup, and a normal MITgcm termination. Build job 290443 created the 11 GB `trajectories_v2.zarr` store with shape $3\times7200\times46\times62\times62$. Independent job 290444 verified finite states, zero land values, exact daily iteration spacing, a bitwise-identical v1 prefix, bitwise raw-MDS extension samples at days 1/1800/3600, and independently reproduced training-only normalizers. Dataset metadata and quality-report SHA-256 values are `ae7be47c4569...78b8cc` and `a6f7e5234ace...cf752`.
>
> Coverage contract `config/trajectories_v2_coverage_audit.json`, SHA-256 `e539ca1bb61a...9197f36`, was frozen only after quality validation and before coverage metrics. Training-only job 290446 then measured temperature/SSH state-RMS effective multipliers of 2.142/2.039 and ten-day-increment multipliers of 2.106/2.253. Thus the predeclared two-times slow-field target passed. The velocity state proxies are not used for this decision because some per-regime integral-time estimates are much less stable between the two decades. Report SHA-256: `7758cfb64e73...18e5`.

Successor-training contract
`config/model_c_successor_training_v1.json`, SHA-256 `3e0a300e73ec...159285`, was frozen before reading any v2 training metric. It holds loss v1, $(24, 16)$ modes, four layers, 10% padding, seed, optimizer exposure, and split-1 records fixed. Phase 1 compares the exact selected width-64 architecture on v2 with a

width-64 candidate changing only Channel-MLP expansion from $0.5C$ to Bire's $4C$. Phase 2 restores width 128, lift/projection width 256, and $4C$ mixing only after both phase-1 reports complete. A prospective training gate requires every U/V/temperature/SSH group in every regime to beat persistence, exact three-step reload, finite 180-day rollout, and normalized state-amplitude ratios in $[0.5, 2]$ at 90 and 180 days. Both tasks in GPU array 290448 completed with exit zero, exact reload, finite rollouts, and every amplitude check passing. The data-only control failed only S1 SSH at 1.103435 times persistence over all 5,020 S1 training pairs. Isolated $4C$ channel mixing reduced that ratio to 1.003573 and improved the aggregate U/V/temperature/SSH ratios by 23.6/22.2/9.4/7.9%, but still fails the exact unrounded gate. Contracted phase-2 job 290597 then completed normally in 1h20m22s. Width 128 passes the complete gate: its aggregate U/V/temperature/SSH ratios are 0.049040/0.121279/0.384816/0.522413, every regime/group is below persistence, and S1 SSH is 0.762683. Exact reload, finite 180-day rollout, and every amplitude check also pass. Report/checkpoint SHA-256 values are `5946515ea81a...68433c`/`2d10e48adc5e...360060`; lightweight phase results are preserved under `outputs/af_fno/C/successor_training_v1/`. This all-group rule is deliberately a training-capacity diagnosis: a dense 46-output model should be able to fit every supervised group. It is not the final validation metric and does not restore SSH as the sole forward-model veto. Width 128 restores the reference's absolute latent, lift/projection, and Channel-MLP widths, but its latent-to-output ratio is still only $128/46 = 2.78$, versus $128/10 = 12.8$ for Bire. A failure would therefore keep group-specific projection or barotropic auxiliary structure scientifically motivated; it would not justify an unbounded width search.

Fresh-validation contract:

> `config/model_c_successor_validation_v2.json`
> SHA-256 `56bb6ec59799...0cf27b`.

It was frozen before any split-2 state metric. It declares seeds 20260723, 20260724, and 20260725 and scores every complete 10–90-day start. Surface speed, SST, and derived surface pressure are primary fields. Persistence, training climatology, and A0 are baselines. The contract also fixes paired regime-stratified block bootstrap and mandatory SSH and streamfunction stability. Contract v1 is preserved but superseded: a metadata-only audit found 180 rather than 181 complete starts per regime (indices 6300–6479); no validation state was read and no scientific criterion changed. GPU array 290673 completed the two missing seeds on split 1. All three seeds pass every prospective training criterion, with worst regime/group ratios 0.762683/0.732270/0.839507, exact reload, and finite amplitude-bounded 180-day rollouts. Frozen fresh-validation GPU job 290738 completed operationally but rejected the configuration scientifically. Surface speed passes every comparison; the three-seed bootstrap-mean SST/surface-pressure RMSE-AUC ratios to persistence are 2.313/2.198. Their mean day-10 ratios are 0.861/0.578 but errors cross persistence around days 20–30 and reach 4.862/4.636 by day 90. This is an autoregressive slow-field failure, not an SSH-only or one-step-capacity failure.

All seeds remain finite, preserve zero land, and pass SSH/streamfunction amplitude and wind-slope checks. The normalized-maximum ceiling 20 is exceeded at 21.169–21.367, but validation truth reaches 21.092; that subgate is non-discriminating and the primary metrics independently reject the model. Contract `config/model_c_rollout_diagnosis_v1.json`, SHA-256 `d0ecee0db2b2...ff7ba`, froze a balanced 10–90-day split-1 audit before any new architecture or loss decision. GPU job 291102 completed in 3m58s with exit zero and every seed reproduced good day-10 slow-field skill followed by failed RMSE-AUC and day-90 skill. Three-seed mean day-10/AUC/day-90 ratios to persistence are 0.809/2.021/4.221 for SST, 0.552/1.867/3.873 for surface pressure, and 0.532/1.844/3.835 for SSH. Apply the predeclared classification: `training_objective_or_checkpoint_gate_mismatch`. This is not merely a validation-only generalization gap and does not justify more MITgcm data. Intermediate winds I1/I2 and inference/perturbation/adjoint truth remain sealed.

Five project-facing validation figures are preserved under `outputs/af_fno/C/successor_validation_` `v1/fresh_validation/`. Their manifest content SHA-256 is `8b24b6af0686...a0845f`; it binds each PNG to the immutable report/array hashes. Three further training-diagnosis figures, a lightweight summary, and a manifest are under `outputs/af_fno/C/rollout_diagnosis_v1/`; diagnosis manifest content SHA-256 is `664e66a0866d...393671`. Neither plotting step read a later archive.

Completed checkpoint, duration, and truncated-unroll audits. The practical guidance in Duruisseaux et al. [2] points to the observed mechanism directly. Pages 62–64 distinguish easy, data-efficient one-step autoregression from its long-horizon distribution shift: training sees true states, whereas inference recursively consumes imperfect predictions and compounds error. Recommended mitigations include short unrolled objectives, pushforward-style training, noise/denoising, contractive behavior, and long-horizon statistical checks. Pages 53–59 also warn against changing learning schedules, normalization, modes, and capacity without first separating optimization, preprocessing, and representation; pages 70–71 note that multi-objective gradients can be badly imbalanced.

Accordingly, another mode/width/data search was not started. The frozen audit contract was:

> `config/model_c_checkpoint_replay_audit_v1.json`
> SHA-256 `494ce9de6688...06682`.

Job 291164 replayed the accepted seed-20260723 training exactly. Architecture, loss v1, optimizer, batch order, seed, and 15,360-step exposure are unchanged. Save the six original late steps 11,520, 13,440, 14,400, 14,880, 15,120, and 15,360; require bitwise equality with the hash-pinned step-14,880 state dictionary and zero numerical difference in the original six-row training history. Score each on the same 540 balanced split-1 starts at days 10–90.

A checkpoint-only correction was admissible only if one exact replay remained finite and land-zero, retains the old every-regime/every-group ten-day gate, beats both persistence and training climatology in 10–90-day RMSE-AUC for surface speed, SST, and surface P/ρ, and keeps SST/P/ρ below both baselines at every lead. The replay completed in 1h19m14s. Step 14,880 is bitwise exact and the original six-row training history has zero numerical difference. No checkpoint passes. Diagnostic best step 14,400 retains ten-day skill but has worst primary RMSE-AUC ratio 2.366 and worst slow-field lead ratio 5.513. Therefore apply the frozen `objective_correction_required` branch.

The first bounded correction is now frozen before training metrics:

> `config/model_c_pushforward_objective_v1.json`
> SHA-256 `406ee5248fb4...bccc2`.

Starting from exact checkpoint 14,400, it preserves the architecture, normalization, and every loss-v1 term. It adds equal, climatology-scaled SST and surface-P/ρ RMSE at alternating day-60/day-90 endpoints. Intermediate recursive states are detached and only the terminal ten-day map receives the auxiliary gradient; this targets inference-state distribution shift with four backpropagated calls per batch rather than a full nine-step graph. The fixed coefficient 0.0025 makes the initial auxiliary contribution comparable to the existing total.

GPU job 291612 completed 1,920 fine-tune steps and evaluated steps 480/960/1,440/1,920 on the same 540 split-1 rollouts. A pass required primary 10–90-day RMSE-AUC below both baselines, SST and surface P/ρ below both at every lead, the old ten-day regime/group gate, finite rollouts, zero land, and exact reload. Step 1,920 is best by every frozen priority and improves worst primary AUC from 2.366 to 1.692 and worst slow-field lead ratio from 5.513 to 3.994, but the gate remains failed. It is not replicated and fresh validation remains closed.

The training-window total and both auxiliary terms decrease at every checkpoint, and the stabilized 2×10^{-5} rate was active only for the final 480 steps. Before changing the loss again, freeze:

`config/model_c_pushforward_duration_v1.json`
SHA-256 `4867fa77d68e...78c79`.

This duration diagnosis replayed the complete 1,920-step path from checkpoint 14,400 with the identical seed, batch order, optimizer, objective, and decay at step 1,440. The replayed step-1,920 state must match the completed job bit for bit before any extension metric is trusted. It then adds 3,840 steps at the already-decayed rate and scores eight checkpoints from total step 2,400 through 5,760 on the same gate. GPU job 291616 completed in 51m31s and the step-1,920 replay is bitwise exact. None of the eight duration checkpoints passes. Diagnostic-best step 5,760 has worst primary RMSE-AUC/slow-field lead ratios 1.597/3.609; at day 90, SST is 2.402/3.609 and surface P/ρ is 2.922/1.727 times persistence/climatology. Further low-rate exposure improves the curves but is not sufficient. Apply the predeclared classification that one-step terminal supervision is structurally insufficient; do not replicate or validate this checkpoint.

The one authorized structural correction is now frozen before its metrics in operationally corrected version 2:

`config/model_c_truncated_unroll_objective_v2.json`
SHA-256 `0c5aabadbe51...92e3c`.

It begins at duration step 5,760 and retains width 128, modes $(24, 16)$, four layers, padding, normalization, loss-v1, the split-1 chronology, batch order, climatology scales, coefficient 0.0025, and the complete 540-record gate. The sole scientific change is gradient horizon. Alternating batches supervise three consecutive endpoints at days 40/50/60 or 70/80/90 and backpropagate through all three ten-day calls; states before the selected window remain detached. This is short truncated backpropagation rather than a full nine-call graph, and directly tests whether the learned operator can control recursive slow-field amplification.

Initial job 291769 stopped before training or scientific metrics because the source validator expected `fine_tune_step`, whereas the duration checkpoint records `total_fine_tune_step`. Version 2 changes no objective or data read: it moves the corrected schema check into preflight. Four focused tests, lint, shell validation, all source hashes, and the training-only sealed-data preflight passed. GPU job 291778 completed normally in 20m01s. All four checkpoints remain finite and land-zero, and diagnostic-best truncated step 1,920 reloads a nine-call rollout bitwise exactly. It fails every scientific gate. Worst primary RMSE-AUC/slow-lead ratios are 1.933/3.701, worse than the duration source's 1.597/3.609. Its SST AUC ratios to persistence/climatology are 1.479/1.933 and its surface-P/ρ ratios are 1.298/0.585; day-90 values are 2.463/3.701 and 2.844/1.681. SST first loses persistence at day 30 and surface pressure at day 60. The completed test therefore rejects short three-call truncation as sufficient and does not authorize replicas or fresh validation. Report/arrays SHA-256 values are `86f7981e7a0a...b5b830`/`0c5222cd6938...d716`; the project-facing figure/CSV/summary are under `outputs/af_fno/C/truncated_unroll_objective_v2/`.

Completed signed-bias diagnosis and completed loss-v3 objective. The evidence now supports slow-field tendency drift more strongly than a general forecast failure. Velocity and the transport streamfunction remain accurate, while SST, SSH, and surface pressure start skillful and then accumulate error against an almost stationary truth. Pure capacity, loss-weight, duration, detached-pushforward, and three-call-gradient variants are exhausted. The training-only frozen-checkpoint audit was

`config/model_c_slow_field_bias_projection_v1.json`
SHA-256 `fd7d39f5c6a4...05428`; job 292064.

It uses every split-1 teacher-forced pair to measure the signed physical ten-day increment error for SST, SSH, and all 15 temperature levels. Nine times each regime-specific bias field is compared with the reference seed's mean day-90 error on the exact 540 starts used by job 291102, using area-weighted pattern correlation, fixed-amplitude explained energy, basin mean, and a basin-mean-plus-five-increment-EOF projection. No checkpoint weight changes.

The same contract applies five post-hoc maps after every frozen model call: raw, zero-wet-area-mean SSH increment, SSH plus per-level temperature-mean conservation, static signed-bias subtraction, and static-bias subtraction plus both constraints. SSH uses cosine-latitude wet-area volume weights. Temperature targets the regime-specific, per-level mean truth tendency measured only on split 1. A training-only scalar AR(1) baseline toward pointwise climatology and 10–720-day SST/SSH anomaly correlations quantify whether persistence is already near the deterministic limit.

Job 292064 completed in 7m09s without opening a held state. Fixed-amplitude 9ϵ explains 33.43% of pooled day-90 SST error energy and 27.21% of SSH, below the predeclared 50% bias-dominance criterion. Basin mean plus the first five training-increment EOFs explain only 4.57%/1.73% of SST/SSH day-90 error energy. Static-bias subtraction plus conservation nevertheless reduces the worst slow-field RMSE-AUC ratio from 1.986 to 1.603, a material 19.25% improvement but just below the frozen 20% alternate threshold; conservation alone changes it to 1.996. Apply the declared `feedback_amplification` branch: static bias is real but not the dominant low-dimensional error, so forecast-age conditioning is primary. The four figures, CSV, and summary are preserved under

`outputs/af_fno/C/slow_field_bias_projection_v1/`.

Immutable report SHA-256 is `e66b263c826c...0f90`; arrays SHA-256 is `4e7f73c2384e...8547`.

The completed training contract is

`config/model_c_rollout_conditioned_loss_v3.json`
SHA-256 `37f1157851ac...5a08`.

It starts from diagnostic-best duration step 5,760, preserves the width-128 one-state architecture, loss-v1 core, split-1 chronology, and 540-record gate, and adds three attributable components. Every model call is followed by the linear zero-area-mean SSH projection and regime/level training-mean temperature-tendency projection. A weight-0.01 per-wet-cell batch-mean signed-increment penalty makes the remaining temperature/SSH bias visible. A weight-0.0025 auxiliary loss cycles over days 10–90; all preceding projected forecast states are detached and exactly one call from the selected forecast age receives this gradient. Thus each update has four differentiable calls—the three-call loss-v1 core plus one conditioned call—without changing the eventual Model-D state interface. Six checkpoints over 2,880 steps at 2×10^{-5} are scored before any replica or held-state read.

GPU job 292293 completed normally in 27m46s and opened no held state. The selected diagnostic checkpoint is loss-v3 step 960 and reloads nine calls bitwise. Exact projection residuals remain below 3.73×10^{-11} for normalized area-mean SSH increment and 1.17×10^{-10} for normalized temperature-mean tendency. All four ten-day groups beat persistence, with aggregate U/V/temperature/SSH ratios 0.0382/0.1058/0.2746/0.3456. The strict 10–90-day gate nevertheless fails: SST RMSE-AUC is 1.119/1.462 and surface-P/ρ is 1.364/0.615 times persistence/climatology; their day-90 ratios are 2.419/3.634 and 3.019/1.785. Worst primary AUC improves from the duration source's 1.597 to 1.462, but worst slow-field lead changes from 3.609 to 3.634. Thus the constraints and forecast-age term improve the primary curve without controlling terminal drift sufficiently; no replica, fresh validation, or automatic loss/capacity variant is authorized. Report/arrays/checkpoint SHA-256 values are `e9af34326c36...7ba11/522fe6145735...edff/ e9f962878cb5...970b`. The two project-facing plots, complete CSV, summary, and hash manifest are under `outputs/af_fno/C/rollout_conditioned_loss_v3/`.

Bire-derived state correction. The archive comparison identifies a difference in state representation more consequential than another layer, mode, or padding change. Bire centres and scales each channel at each grid point over training time and predicts the normalized future state directly. The present Model C instead uses one scalar mean/scale per channel and predicts a small residual increment. Its slow fields therefore occupy poorly conditioned nearly identity coordinates: a small systematic feedback error is inexpensive at one call but accumulates at every recursive call.

The primary representation contract is frozen before its training or validation metrics:

> `config/model_c_anomaly_direct_v1.json`
> SHA-256 `fa38b41578f5...8acea`.

It computes one wind-independent pointwise mean and population standard deviation from all 15,120 split-1 S0–S2 snapshots. Pooling regimes keeps the normalization smooth with respect to wind amplitude for the later derivative study. Each channel's wet point scale is floored at its fifth percentile, with a 10^{-6} absolute guard. This changes exactly 5% of wet points per channel; a floor equal to 1% of the old scalar scale was rejected during preflight because it would floor 69% of SST points. The exact mean, raw-scale, and floored-scale SHA-256 values are `36489318e23f...a649`, `ca1cb352d25b...93d7`, and `b08f340a70f1...a83f`.

The FNO output is the next pointwise-normalized anomaly state; it is never added to the current state. A zero dynamic output therefore maps to the pooled pointwise training climatology. Width 128, four blocks, modes $(24, 16)$, 10% padding, lift/projection width 256, the $4C$ Channel MLP, local 3×3 branch, loss-v1 three-step core, optimizer, seed, and batch order remain fixed. Seven checkpoints from steps 3,840–15,360 are selected only on the same 540 balanced split-1 rollouts and strict 10–90-day gate. After that selection, the same prospectively fixed 15 S2 members are characterized through day 200. The required deliverable is

> `outputs/af_fno/C/anomaly_direct_v1/`
> `model_c_bire_figure4_dt10_rmse_0_200_days.png`,

along with normalization, checkpoint-selection, training-curve, streamfunction, and unclipped full-range plots. LayerNorm/dropout, a three-block/no-padding control, and group-specific heads remain subsequent bounded tests rather than simultaneous changes. Eight focused and inherited figure tests, lint, shell validation, source hashes, normalizer preflight, and a production-grid three-step differentiable smoke test pass. Inference, intermediate-wind, response, and adjoint states remain sealed.

Completed representation result, 28 July 2026. V100 job 303447 completed normally in 1h29m32s and selected optimizer step 14,400 using split 1 only. Its strict training-only gate passes: the worst primary RMSE-AUC ratio to either baseline is 0.394 and the worst SST/surface-P/ρ single-lead ratio is 0.519. Aggregate ten-day U/V/temperature/SSH ratios to persistence are 0.0820/0.1905/0.2345/0.2645. On the prospectively fixed S2 members, 10–90-day surface-speed/SST/surface-P/ρ RMSE-AUC ratios to persistence are 0.222/0.206/0.601; the corresponding ratios to climatology are 0.313/0.258/0.214. At day 200 their RMSEs remain bounded at 0.00212 $\mathrm{m\,s^{-1}}$, 0.0288 °C, and 0.0552 $\mathrm{m^2\,s^{-2}}$, compared with the loss-v3 map's approximate 0.058, 4.50, and 3.13. Surface pressure no longer runs away, although its day-200 error is 2.18 times the exceptionally small persistence error; it remains 0.768 times climatology. The requested Figure 4 and all five companion plots are saved under the contracted project output root.

This is evidence for the combined representation change, not yet a claim that every random seed or the complete 360-day physical gate passes. Before changing layers, padding, regularization, or heads, the independent-seed contract

`config/model_c_anomaly_direct_replication_v1.json`
SHA-256 `dd7e0f92ef50...d8325f`

freezes replicas 20260724/20260725. Only initialization and seed-driven batch order may change. Each checkpoint is again selected on split 1 before repeating the already fixed 15-member S2 characterization.

Completed replication result, 29 July 2026. Array 303640 completed both tasks normally, and all three seeds pass every frozen gate. Their mean six-ratio ranking scores are 0.30227, 0.24717, and 0.23184 for 20260723/20260724/20260725, so the predeclared median is seed 20260724 at step 13,440. Its day-200 speed/SST/surface-P/ρ RMSE is 0.00242/0.0262/0.0223, below both persistence and climatology. The sealed summary, CSV, manifest, and comparison plot are under `outputs/af_fno/C/anomaly_direct_replication_v1/`.

The complete held-inference contract was frozen before opening split 3:

`config/model_c_anomaly_direct_forward_v1.json`
SHA-256 `d11029fce3aa...c2f6c`.

It fixes 45 starts wholly inside the fresh trajectory-v2 inference block, 36 ten-day calls, the median checkpoint, persistence, regime climatology, training-only damped persistence, and A0 on identical starts. The separate adjoint-readiness gate requires all three primary fields at days 10 and 30 to beat every baseline with positive ACC; the long gate adds 10–90-day AUC, day-360 boundedness/drift, resolved spectra, and wind-response slope. The forcing-switch sign remains reported but non-vetoing until matching MITGCM truth exists. This job may open only the declared inference states; intermediate-wind, response, and adjoint archives remain sealed.

Completed held-inference result, 29 July 2026. V100 job 303993 completed normally in 3m00s. The separate 10/30-day adjoint-readiness determination passes every primary-field comparison. At day 30 the model RMSE for surface speed, SST, and surface-P/ρ is 0.000781 $\mathrm{m\,s^{-1}}$, 0.00798 °C, and 0.00524 $\mathrm{m^2\,s^{-2}}$, with ACC 0.9767/0.9778/0.9977; each is below persistence, climatology, damped persistence, and same-start A0. The primary 10–90-day RMSE-AUC also beats all four baselines. The one-year rollout is finite, exactly zero on land, and bounded at a maximum normalized magnitude of 15.81; final U/V/temperature/SSH mean biases are only 0.00676/0.00306/0.000285/0.00653 training standard deviations. The streamfunction wind-slope ratio is 1.180 and the provisional forcing-switch sign is correct.

The combined gate nevertheless remains a frozen *failure*. Five of seven day-360 spectral ratios pass, but the modewise medians for mid-depth and bottom `PHIHYD` are 22.29 and 9.63, above the factor-four bound. Consequently the complete adjoint campaign remains closed even though the short-horizon scientific determination is retained as a pass. The complete metrics, arrays, scorecard, readiness plot, spectra, maps, snapshots, and stability figures are under `outputs/af_fno/C/anomaly_direct_forward_v1/`.

The immutable posthoc diagnosis `outputs/af_fno/C/anomaly_direct_forward_spectral_diagnosis_v1/` does not reinterpret that failure. It shows that total integrated day-360 energy is 1.029 times truth at mid-depth and 0.947 at the bottom, while modes $k \geq 10$ contain only 0.0159% and 0.0193% of the respective model energy (truth shares 2.24×10^{-6} and 5.26×10^{-6}). Thus the failure is a real but small high-wavenumber tail rather than basin-scale drift or an unstable attractor. Before changing weights, the training-only attribution contract

```
config/model_c_anomaly_direct_training_spectral_attribution_v1.json
SHA-256 afb64d5a0f3a...d0246
```

fixes all three accepted seeds, 36 split-1 starts, all 15 pressure levels, and 360-day rollouts. It will establish seed consistency, depth, and onset without reading validation, inference, response, or adjoint state; only a seed-consistent split-1 tail authorizes one prospective spectral-regularization fine-tune.

Rejected attribution implementation, 29 July 2026. Job 304125 completed normally in 1m07s, but its scientific output is invalid. The runner advanced the model by one ten-day call while gathering truth at daily offset $t + s$, not $t + 10s$. The impossible 10–90-day primary RMSE ratios 3.48–4.48 times persistence exposed the mismatch. Its output is retained only as rejected provenance and cannot authorize a loss change. The corrected contract

```
config/model_c_anomaly_direct_training_spectral_attribution_v2.json
SHA-256 29c344310e50...a3de1
```

pins a stride of ten daily snapshots per model call, verifies the inclusive t through $t + 360$ window stays wholly in split 1, and replaces the invalid starts with 12 fixed admissible times. Six focused tests, lint, shell validation, source hashes, and production-data preflight pass. No regularization or later archive read is authorized until this corrected attribution completes. V100 job 304177 completed normally in 1m11s and validly classifies a seed-consistent training-split tail. Across seeds 20260723–20260725, day-360 mid-level ratios are 12.25–13.54 and bottom ratios are 4.96–5.69; failure begins at days 170–190 and 310–330, respectively. Total mid/bottom pressure energy remains 1.02–1.07 times truth, while every 10–90-day primary ratio remains only 0.19–0.37 times persistence. Report content SHA-256 is `f3cb1bd6de11...141927`, and the immutable package is under `outputs/af_fno/C/anomaly_direct_training_spectral_attribution_v2/`.

Single authorized spectral fine-tune, 29 July 2026. Contract

```
config/model_c_anomaly_direct_deep_pressure_spectral_regularization_v1.json
SHA-256 4f3377382dc0...acef37
```

starts from median seed 20260724 at step 13,440 and changes only one term: a weight 10^{-3} differentiable complex-Fourier PHIHYD error over levels 7–14 and radial modes 10–30 on the existing 10/20/30-day rollout. It fixes 1,920 constant-2×10^{-5} fine-tune steps and five checkpoints. The same 36 corrected split-1 records select a candidate only if both mid and bottom spectra stay within factor four at every lead through day 360, total energy remains 0.8–1.25 times truth, every primary remains better than persistence, and no primary degrades more than 10% from the source. Four focused tests, lint, shell validation, source locks, and production-data preflight pass. V100 job 304324 completed normally in 10m25s. The term substantially helps but does not pass the prospective rule: at fine-tune step 1,920 the day-360 mid/bottom ratios fall from 13.37/5.57 to 6.19/2.97, the bottom tail never crosses four, and mid-tail onset moves from day 180 to day 280. Integrated mid/bottom energy remains 1.006/0.980 times truth and all primary 10–90-day ratios improve or remain protected. Because the mid-level ratio still exceeds four, no fine-tune checkpoint is promoted; the selected model remains original seed 20260724 at step 13,440. Report/manifest content SHA-256 is `8065faafeecf...ddb75/ 280a863a5892...f3a8`, with JSON, arrays, and two plots under `outputs/af_fno/C/anomaly_direct_deep_pressure_spectral_regularization_v1/`. No second spectral weight or duration is authorized, and validation and later archives remain sealed.

Bire regularization factorial, 29 July 2026. Contract

```
config/model_c_anomaly_direct_bire_regularization_controls_v1.json
SHA-256 dd9089e9675a...0f827
```

implements the next originally ordered Bire-derived control without extending the failed spectral loss. Three from-scratch seed-20260724 arms isolate pointwise channel LayerNorm after spectral and Channel-MLP mixing, archived Channel-MLP dropout 0.5, and their interaction. The unregularized seed-20260724/step-13,440 model is the fixed reference. Every arm retains the pointwise anomaly/direct-state representation, width 128, four layers, modes 24×16, padding 0.1, loss v1, data order, optimizer, and 15,360-step budget. Seven checkpoints per arm are scored only on the corrected 36-member split-1 protocol through day 360 with the same factor-four, integrated-energy, persistence, and 10%-degradation guards. Four focused tests, lint, shell validation, all-arm production preflight, and finite 62-by-62 forward smoke tests pass. No validation, held-inference, long-term, response, or adjoint state may be read. All tasks of V100 array 304376 completed normally in 1h09m29s–1h23m33s. No arm passes the frozen gate. At its best diagnostic checkpoint, LayerNorm gives day-360 mid/bottom modewise ratios 9.74/4.92 and a worst primary degradation of 18.6%; dropout gives 37.59/15.04 and 57.4%, and the combined arm gives 19.51/8.27 and 63.2%. Thus LayerNorm helps the source mid-level ratio 13.37 but does not reach factor four or preserve primary skill within 10%; dropout is rejected. Source-locked aggregate contract SHA-256 `c977c196b487...eaea`, report-content SHA-256 `9f7b7cbbf5dc...5527`, and manifest-content SHA-256 `fb6e622fef1f...53b2` retain original seed 20260724/step 13,440. The aggregate CSV and cross-arm plot are under `outputs/af_fno/C/anomaly_direct_bire_regularization_controls_v1/aggregate/`.

Bire three-layer/no-padding control, 29 July 2026. The negative regularization aggregate activates the next predeclared comparison:

> `config/model_c_anomaly_direct_three_layer_no_padding_v1.json`
> SHA-256 `7d405cfb8307...fd20`

changes only four to three FNO blocks and domain padding 0.1 to none. It retains seed 20260724, pointwise anomaly/direct-state coordinates, width 128, modes 24×16, lift/projection 256, Channel MLP $4C$ without dropout, no block normalization, loss v1, optimizer, 15,360-step budget, seven checkpoints, and the corrected 36-member split-1 day-360 gate. This faithful combined Bire control cannot separately attribute layer count from padding. Nine focused tests, lint, shell validation, immutable-source preflight, and a finite 62-by-62 forward smoke test pass; the control has 11,174,614 trainable parameters. Validation, held-inference, long-term, response, and adjoint archives remain sealed.

V100 job 304526 completed all 15,360 training steps and wrote all seven source-contract checkpoints, then failed operationally before its first metric. The inherited attribution evaluator attempted to reconstruct the intentional three-layer checkpoint with the old four-layer-only architecture dataclass. No scientific result is assigned. The checkpoint SHA-256 values, failed log SHA-256 `7488ddfc467e...c0ce`, and incomplete temporary directories are frozen by the evaluation-only recovery contract

> `config/model_c_anomaly_direct_three_layer_no_padding_recovery_v2.json`
> SHA-256 `9232cea2ae58...165b`.

The recovery performs zero training steps and changes no weight; it only loads the same seven checkpoint bytes with the declared architecture, runs the unchanged 36-record split-1 scoring, and atomically publishes the missing report, arrays, plot, README, and manifest. Eleven focused tests, lint, shell validation, and production preflight pass. All later archives remain sealed. Recovery job 304597 completed normally in 2m41s and supplied the missing scientific decision. No checkpoint passes. Diagnostic-best step 14,400 has day-360 mid/bottom modewise ratios 12.51/5.84, failure onset at days 210/310, and integrated-energy ratios 1.032/1.011. Its surface-speed, SST, and surface-P/ρ ratios to persistence are 0.225/0.211/0.386, but its worst primary degradation relative to the selected source is 1.131, outside the 1.10 guard. Thus three layers and no padding delay the mid-level tail from day 180 to day 210 and slightly lower its terminal ratio, but satisfy neither the factor-four rule nor primary protection. The original seed-20260724/step-13,440 Model C remains selected. Report/manifest content SHA-256 is `a429db0f64bb...58416/ 152190e6d06f...14539`, with arrays and the selection plot under `outputs/af_fno/C/anomaly_direct_three_layer_no_padding_v1/three_layer_no_padding/`.

Group-specific projection-head control, 29 July 2026. The negative architecture control activates one training-only capacity test:

> `config/model_c_anomaly_direct_group_heads_v1.json`
> SHA-256 `932afdd1b6b5...b1c6`.

The source four-block, width-128, 24 × 16-mode, 10%-padded FNO trunk is unchanged. Its 128-channel output feeds independent 128 → 256 → 15/15/15/1 pointwise projection heads for U, V, temperature, and SSH, with matching independent 3 × 3 local paths. No LayerNorm or dropout is added. The pointwise anomaly/direct-state representation, seed, data order, loss v1, optimizer, 15,360-step schedule, seven checkpoints, and corrected 36-record split-1 gate remain fixed. A passing checkpoint must keep both mid/bottom spectra below four at every lead through day 360, bound terminal integrated energy to 0.8–1.25, beat persistence for every primary field, and stay within 10% of the selected source's primary skill. Three focused tests, lint, shell validation, immutable-source preflight, and a finite native-grid forward test pass. The 15,026,902-parameter control reads no validation, held-inference, long-term, response, or adjoint state. Failure retains the original Model C and closes this architecture-control branch; no head-width or grouping search is automatic.

V100 job 304708 completed normally in 1h30m46s, but no checkpoint passes. Diagnostic-best step 14,880 preserves the primary guard: its 10–90-day speed/SST/surface-P/ρ ratios to persistence are 0.211/0.202/0.360 and its worst source-relative degradation is 1.055. It improves day-360 mid/bottom modewise ratios from the source's 13.37/5.57 to 9.21/4.40, delays their failure onset from days 180/310 to 220/350, and keeps integrated energy at 1.004/0.995. Nevertheless both levels still exceed four, so the grouped heads are diagnostic only and the original Model C remains selected. Report/manifest content SHA-256 is `fa25da048b5d...0f3b6/8e9c6a867727...5488e`; all artifacts are under `outputs/af_fno/C/anomaly_direct_group_heads_v1/group_specific_heads/`.

Bire long-term skill and learned-physics scope. A replicated candidate must still undergo a Bire-section-5.3-style approximately 2,000-day ensemble test. Beyond decorrelation this is a bounded statistical test of climatological mean and variance, drift, spatial spectra/scales, and mean circulation, not a claim of trajectory RMSE superiority. A Bire-section-6 learned-physics test is also required: initial-state and changed-wind sensitivity must have the correct sign, amplitude, spatial pattern, and time evolution. The existing one-year boundedness and scalar wind-slope checks are promising precursors, not substitutes. Spatial response and forward/adjoint archives remain sealed until the original combined forward gate passes.

Frozen S0 control-wind figure suite, 29 July 2026. The requested paper-aligned evaluation uses only the control wind, $\tau_0 = 0.1\ \mathrm{N\,m^{-2}}$, and the selected median replicate (seed 20260724, step 13,440). Contract `config/model_c_bire_s0_long_truth_v1.json`, SHA-256 `dc204636660d...d75ea5`, prospectively draws 15 unique initial conditions without replacement from the fresh late S0 inference block (indices 6660–7199) using NumPy seed 20260729. The draw is reused for every ensemble panel; start 6913 is fixed for the single-member panels. All maps use a ten-day prediction interval.

The existing archive ends at index 7199, so an exact six-year continuation from the immutable year-120 S0 pickup supplies unseen truth through index 9359. This is sufficient for the largest selected target, index 9199, and is explicitly evaluation-only: it is never appended to training data. CPU job 304735 completed normally in 6m59s from iterations 3,110,400 to 3,265,920. It produced all 2,160 paired daily `dynState/surfState` records, the final pickup, and `STOP NORMAL END` on all four ranks. Result/manifest SHA-256 is `9ebd446ff6f3...99925/e15530c67dbe...41c6f`.

Evaluation contract `config/model_c_bire_s0_figures_v1.json`, SHA-256 `6afa41cfaaf8...616030`, source-locks that truth, both comparison checkpoints, pointwise normalization, dataset, diagnostics, output paths, and the following six figures before any new metric. One frozen GPU evaluation produces exactly six figures: Figure-3-style streamfunction maps at days 0–40; Figure-4-style 15-member speed, surface-P/ρ, and SST RMSE through day 200; one-member streamfunction/SST RMSE; four-field ACC through day 200; streamfunction maps at days 60 and 2,000; and 15-member RMSE through day 2,000. Climatology is the S0 pointwise time mean over split-1 training snapshots only, while persistence holds each member's initial physical field fixed. Figure 6 compares the prior residual rollout-conditioned Model C with the selected pointwise-anomaly direct-state Model C. Because the latter was trained from scratch, this is labeled an architecture-direction comparison, not falsely described as Bire's literal pretrained-to-finetuned pair. Four focused tests, lint, shell validation, complete-truth preflight, and native-grid one-step reloads of both models pass.

GPU job 304736 completed normally in 1m07s and generated the prospectively fixed numerical archive and all six figures. At day 200 the selected model beats both baselines for every Figure-4 field: model/persistence/climatology mean RMSE is 0.00270/0.00457/0.00351 $\mathrm{m\,s^{-1}}$ for surface speed, 0.0151/0.0198/0.0625 m for surface P/ρ, and 0.0241/0.0488/0.0391 °C for SST. Its day-200 ACC for surface U, surface V, P/ρ, and SST is 0.784/0.887/0.972/0.833, versus 0.107/0.033/0.003/−0.036 for the prior residual model. The anomaly/direct-state representation is therefore the best Model C direction for deterministic 10–200-day skill.

It does not pass Bire-style long-term stability. Although every member stays finite, day-2,000 model RMSE reaches 0.426 $\mathrm{m\,s^{-1}}$, 2.593 $\mathrm{m^2\,s^{-2}}$, and 5.339 °C for speed, surface P/ρ, and SST, or 81.9/67.7/87.1 times persistence. For the fixed spatial member, streamfunction RMSE grows from 0.294 Sv at day 60 to 96.2 Sv at day 2,000, with correlation falling from 0.9996 to −0.183. The 10th–90th percentile bands show the runaway across the ensemble rather than in one outlier. Retain this checkpoint for the 10–30-day forward/adjoint purpose, but do not claim a bounded long-term emulator; a separate statistical-stability correction and a spatial changed-wind learned-physics test remain required. The complete immutable package is under `outputs/af_fno/C/anomaly_direct_bire_s0_inference_v1/`; arrays/report SHA-256 is `30abd95810d3...3c7/ 32ac2e051726...bd18`, and manifest-content SHA-256 is `593dd43de772...57d1`.

Frozen S0 boundary/checkpoint diagnosis, 29 July 2026. Figure 7 is diagnostic rather than merely illustrative: for the fixed member, the four-cell wet-boundary union contains 71.3% of day-2,000 streamfunction squared error, the western four columns 45.2%, and the northern four rows 31.5%. Because streamfunction is reconstructed by integrating all 15 predicted U levels from south to north, this implicates a boundary-localized barotropic-transport mode, while the cumulative integral can amplify a localized U bias into a broad streamfunction stripe.

Contract `config/model_c_bire_s0_boundary_checkpoint_v1.json`, SHA-256 `e859999279a7...d2c0`, freezes a zero-retraining classification before reading the corresponding 15-member/checkpoint metrics. It evaluates the seven existing seed-20260724 checkpoints at optimizer steps 3,840/7,680/11,520/13,440/14,400/14,880/15,3 on the exact job-304736 S0 starts and truth through day 2,000. For depth-integrated Q_x, Q_y, and U-derived streamfunction it records regional RMSE, boundary squared-error fraction, post-day-200 growth timing, normalized-amplitude thresholds, and day-2,000 spatial-pattern/sign agreement. Five fixed plots separate checkpoint age from boundary versus interior growth. This diagnostic may neither retrain nor reselect a checkpoint. Four tests, lint, shell validation, all checkpoint/artifact hashes, complete-truth preflight, and finite land-zero one-step reloads of every checkpoint pass. Planned results are isolated under `outputs/af_fno/C/anomaly_direct_bire_s0_boundary_checkpoint_v1/`.

Completed S0 boundary/checkpoint diagnosis, 29 July 2026. V100 job 304740 completed normally in 1m46s. All seven checkpoints remain finite but cross mean maximum normalized amplitude 20 by day 880 or earlier; the three earlier checkpoints cross on days 410/580/780, versus day 880 for the selected step 13,440. The selected checkpoint also crosses 40/100/200 on days 1,210/1,610/1,840. Thus shorter training does not avoid the day-2,000 runaway, and optimizer age is rejected as its sufficient cause. This is descriptive confirmation of the frozen checkpoint, not retrospective selection.

The four-cell boundary union is only 896 of 3,600 wet cells (24.9%) but, for selected step 13,440 at day 2,000, contains 72.8%, 85.2%, and 72.7% of total Q_x, Q_y, and streamfunction squared error. Boundary versus interior ensemble-mean RMSE is 147.4 versus 51.1 $\mathrm{m^2\,s^{-1}}$ for Q_x, 100.6 versus 23.8 $\mathrm{m^2\,s^{-1}}$ for Q_y, and 192.7 versus 63.3 Sv for streamfunction. The day-2,000 error pattern is also ensemble-coherent: member-to-ensemble-mean spatial cosine is 0.783/0.649/0.902 and hotspot sign agreement is 93.2/88.5/97.1% for $Q_x/Q_y/\psi$.

The timing evidence is more specific than the original Figure 7 inference. For the selected model, boundary/interior day-200-relative $2\times/5\times/10\times$ crossings are 360/620/860 versus 340/570/840 days for Q_x, 360/600/790 versus 340/550/820 for Q_y, and 660/920/1,260 versus 320/590/920 for streamfunction. Boundary error is larger and carries a disproportionate share throughout the rollout, but it does not consistently grow first. Classify a systematic, boundary-amplified unstable transport mode rather than a single-member accident or a demonstrated boundary-to-interior propagation cascade. Report/summary/arrays SHA-256 is `bc62501a18f7...2fc67/c6c069e52ee5...fef50e/ d37d570337a3...b9942ae`; all five plots and the complete numerical package are stored in the path above.

Post-hoc stability-instrument correction, 29 July 2026. Contract `config/model_c_s0_stability_instrument_v1.json`, SHA-256 `ab6d597843b1...fc5be7`, re-expresses only the immutable job-304736 arrays; it performs no rollout, reads no new state, trains no weight, and cannot select a checkpoint. Ordinary least squares of log ensemble-mean RMSE against ten-day call number, with 10,000 fixed-seed member-bootstrap replicates, shows days-300–600 gain 1.0332 [1.0297, 1.0369] for surface speed, 1.0397 [1.0391, 1.0401] for surface P/ρ, and 1.0456 [1.0424, 1.0485] for SST. On days 1,700–2,000 the corresponding gain remains 1.0321/1.0240/1.0227. Maximum absolute normalized state grows by 1.0252 per call on days 300–600, an approximately 401-day e-folding, from ensemble means 3.49/3.75/5.57/322.39 at days 0/200/300/2,000.

This confirms the central attached-analysis diagnosis: the former day-200 “finite and inside the plotting axis” requirement ended before a persistent supercritical rollout became visually obvious. It does *not* identify the fitted RMSE gain with a Jacobian spectral radius; deterministic chaotic error can grow before saturation. Future acceptance therefore separates 10–90-day trajectory skill from at least 200-call statistical stability. The latter must test post-decorrelation saturation, normalized-amplitude envelopes, time-averaged mean/variance and low/mid/high spectral energy, and streamfunction extrema/mean circulation against MITgcm. A follow-up tangent audit must report dominant singular gain and finite-time tangent growth unless an actual eigenvalue calculation justifies spectral-radius language.

LayerNorm is not presumed to solve the problem. It normalizes hidden channels within a block but does not bound residual/local/projection paths or the full autoregressive map. Moreover, the already completed LayerNorm-only arm missed the day-360 protected gate at mid/bottom spectral ratios 9.74/4.92 and 18.6% worst primary degradation. A zero-retraining day-2,000 comparison can still test its causal effect, but cannot promote it without the short-skill and statistical-stability gates.

Frozen zero-retraining stability/tangent comparison, 29 July 2026. Contract `config/model_c_s0_stability_tangent_comparison_v1.json`, SHA-256 `5d32d472509a...bca4`, fixes the next causal diagnostic before its numerical results are read. On the exact 15 job-304736 S0 starts it rolls three immutable maps for 200 ten-day calls: selected pointwise-anomaly direct-state Model C at step 13,440, the already-trained LayerNorm diagnostic at step 14,400, and the prior residual Model C. It compares speed, surface P/ρ, SST, and streamfunction means/variances; normalized-amplitude envelopes; RMSE growth in three frozen windows; and radial spectra at days 200/500/1,000/1,500/2,000 against the completed job-304735 MITgcm truth and training-only S0 climatology.

For the two direct-state maps only, the same job estimates dominant one-step singular gain and ten-call tangent geometric gain in their shared pointwise-normalized coordinates. Three fixed training-only S0 states and full, $k = 1$–3, $k = 4$–9, and $k = 10$–30 perturbation bands are used. These quantities are not called spectral radii, the prior residual map is excluded from the coordinate-dependent tangent ranking, and no model is trained, selected, or promoted. Three focused tests, lint, shell validation, all source/artifact hashes, the complete long-truth check, and CPU one-step finite/land-zero reloads of all three models pass. Results will be isolated under `outputs/af_fno/C/anomaly_direct_s0_stability_tangent_comparison_v1/`.

Evaluation-only recovery, 29 July 2026. V100 job 304750 exited 1 after 1m42s. Its complete preflight passed, but an explosive comparator overflowed float32 squared-error and spectral reductions; the subsequent log-growth fit correctly refused the non-finite series. The transactional writer published no result package, so this is an operational reduction failure rather than a completed comparison.

Recovery contract `config/model_c_s0_stability_tangent_recovery_v2.json`, SHA-256 `8bc5c0937d0e...e1ac`, binds the failed log, original contract, all unchanged checkpoints, starts, truth, leads, fields, spectra, and tangent states. It changes only the diagnostic arithmetic: model stepping remains float32, physical reductions use float64, the first non-finite model-state lead is recorded per member, and a requested growth window is censored at the first incomplete all-member sample. Four focused tests include large-amplitude overflow, original-versus-float64 diagnostic agreement, and censored-fit behavior. Lint, shell validation, immutable hashes, original preflight, and all-model one-step smoke tests pass. No retraining, checkpoint selection, or promotion is allowed; recovered outputs will be isolated under `outputs/af_fno/C/anomaly_direct_s0_stability_tangent_recovery_v2/`.

Completed recovery and causal result, 29 July 2026. Recovery job 304751 completed normally in 2m12s. All three maps remain numerically finite through day 2,000; the failed-run overflow came from squaring an enormous but finite prior-residual state. Thus finiteness is rejected as a stability criterion. The prior residual comparator is not contracting: its day-2,000 speed/P/ρ/SST RMSE is 3.32×10^{25}, 6.76×10^{26}, and 1.41×10^{27}, and its normalized-amplitude ratio to truth is 1.94×10^{26}. Its fitted gain is approximately 1.38–1.42 per ten-day call, an e-folding time near 29–31 days.

LayerNorm is a consequential but insufficient stabilizer. Its day-2,000 speed/P/ρ/SST RMSE is 0.0217/0.5525/0.4218, versus 0.4260/2.5933/5.3391 for selected Model C; equivalently it retains only 5.1/21.3/7.9% of selected error. Nevertheless, it is still 5.54/6.48/10.12 times climatology, its normalized-amplitude ratio is 4.01, and every frozen RMSE-growth window remains above one. In the late days-1,700–2,000 window, speed/P/ρ/SST gains are 1.00330/1.00541/1.00646 per call. Day-2,000 high-k power is still 25.5/38.9/77.0/31.5 times truth for speed/P/ρ/SST/streamfunction, although far below selected Model C's 3,335/9,240/13,237/10,207.

The tangent audit rejects a uniform-contraction explanation. Averaged over the three training states, selected versus LayerNorm full-field dominant one-step singular gain is 4.04 versus 3.66, but ten-call geometric gain is 1.020 versus 1.117. LayerNorm lowers the high-k one-step/ten-call values from 3.65/0.992 to 2.36/0.964, while increasing mid-k values from 3.34/1.085 to 4.07/1.130. Its improved nonlinear long rollout is therefore associated most clearly with high-k suppression and altered off-manifold dynamics, not contraction of every local band. It also worsens short S0 error: at day 30 its speed/P/ρ/SST RMSE is 1.09/1.37/1.09 times selected Model C. Do not promote it.

All ten manifest artifacts verify. Arrays/report/summary SHA-256 is `95e0a42bc437...01175/684815b7882b...7b84/bdd7c1a06c0d...cc25`; report-content and manifest-content SHA-256 is `05e92283b2a0...00dd/30a5288053ef...c4e` The next bounded correction should preserve selected Model C's short skill while testing whether controlled high-k suppression or a gated normalization path prevents the off-manifold cascade. A blanket Jacobian-norm penalty is not justified by this audit.

Frozen high-k causal control, 29 July 2026. Contract `config/model_c_s0_highk_damping_` `control_v1.json`, SHA-256 `cf2f1c179d36...9554`, tests a zero-retraining postprocessor before another architecture or loss change. After each selected direct-state call, it blends the normalized anomaly with a reflected-boundary channelwise 3×3 binomial smoothing at strength $\alpha = 0, 0.02, 0.05, 0.10, 0.20$. Reflection is applied only on the 60×60 wet rectangle and land is reset exactly to zero.

Strength selection is restricted to ten fixed S0 starts whose complete day-0–1,000 windows lie in split 1. The smallest nonzero strength must keep every day-10/30/90 speed/P/ρ/SST RMSE within 5% of the unfiltered source and reduce, at day 1,000, worst RMSE/climatology and normalized amplitude to at most 80% of source and worst high-k power to at most 50%. Only a passing training choice may open the fixed 15-member S0 day-2,000 inference characterization. That characterization cannot promote a checkpoint; it tests whether high-k suppression is causal and sufficient.

Three tests, lint, shell validation, all hashes, exact zero-strength identity, complete split-1 windows, and a finite land-zero native-grid filtered forward pass succeed. No parameter is trained or selected.

Completed high-k control, 29 July 2026. V100 job 304753 completed normally in 1m07s, and every published artifact verifies. No nonzero strength passes the frozen training gate. The weakest intervention, $\alpha = 0.02$, does pass all three long-horizon reduction requirements: at day 1,000 its worst RMSE/climatology, normalized amplitude, and worst high-k power are 0.7768, 0.7742, and 0.4448 times the unfiltered source. It nevertheless raises the worst protected short RMSE to 1.0658 times source, driven by day-90 SST; day-90 speed is also 1.0612 times source. Stronger filtering further reduces the long tail but rapidly damages short skill (1.5000/2.2640/3.3406 worst short ratios for $\alpha = 0.05/0.10/0.20$).

The conditional 15-member inference archive therefore remains sealed, no strength is selected, and no checkpoint is promoted. The intervention gives causal evidence that high-k content participates in the off-manifold growth, because its removal jointly lowers high-k power, amplitude, and long RMSE. It also shows that fixed isotropic post-step smoothing is not sufficient: it cannot distinguish unstable content from short-range useful speed/SST scales. Do not weaken the prospective guard or tune another fixed α. The next bounded model change should learn or gate the normalization/spectral correction during training while retaining the unfiltered short forecast as an explicit reference.

Arrays/report/summary SHA-256 is `88eac7163328...9d/8ca08f114763...a66/ 3c100081bd10...806`; report-content and manifest-content SHA-256 is `856612ebb748...e11/248c1b2e9a8a...057`. Results are isolated under `outputs/af_fno/C/anomaly_direct_s0_highk_damping_control_v1/`.

Frozen reflected spectral-gate control, 29 July 2026. The bounded follow-up is `config/model_c_s0_reflected_spectral_gate_control_v1.json`, SHA-256 `69b0100aae27...b8e7`. It changes no weight and retains the same ten training-only S0 day-1,000 trajectories, protected leads, long gates, and conditional inference seal. The only intervention change is scale selectivity. A two-dimensional even extension of the 60×60 wet rectangle gives a reflected spectral basis; its transfer is exactly one for radial $k \leq 8$, follows a cosine transition for $8 < k < 10$, and is $1 - \alpha$ for $k \geq 10$, with $\alpha = 0, 0.25, 0.50, 0.75, 1.00$. Land is reset to zero after every call.

This is not another tuning pass over the rejected binomial strength. It asks whether useful short scales and unstable high-k content can be separated by the already diagnosed $k = 10$ boundary. The smallest passing nonzero strength alone may open the unchanged fixed 15-member S0 day-2,000 characterization; the control still cannot select or promote a checkpoint. Six combined old/new focused tests, lint, shell validation, immutable hashes, exact zero identity, constant/smooth-mode preservation, checkerboard attenuation, and a finite land-zero native-grid preflight pass. Planned outputs are isolated under `outputs/af_fno/C/anomaly_direct_s0_reflected_spectral_gate_control_v1/`.

Completed reflected spectral-gate control, 29 July 2026. V100 job 304754 completed normally in 1m06s, and all nine manifest-listed artifacts verify. No nonzero strength passes; fixed inference therefore remains sealed. At the weakest $\alpha = 0.25$, day-1,000 normalized amplitude, worst high-k power, and worst

RMSE/climatology fall to 0.4365/0.0638/0.6178 of the unfiltered source. Despite removing 93.6% of the source high-band power, this state remains 3.65 times truth in normalized amplitude and 4.41 times climatology in worst RMSE. Short skill is destroyed: the worst ratio is 6.49, and SST at days 10/30/90 is 5.30/6.49/5.94 times source. Larger strengths raise the worst short ratio to 10.45/15.63/20.82 without restoring a climate-like day-1,000 state.

This closes fixed postprocessing as the immediate stability direction. High-k excess is a coupled symptom and amplifier, but it is neither the sole source nor cleanly separable from useful short dynamics. The next highest-value hypothesis is the unresolved output bottleneck: width 128 serves 46 outputs here ($128/46 = 2.78$) versus ten in Bire ($128/10 = 12.8$). A reduced ten-output, Bire-facing forward baseline should be established before resuming the complete adjoint campaign. Its long-horizon requirement must be bounded, statistically plausible day-2,000 behavior, not trajectory RMSE below persistence after decorrelation.

Contract/arrays/report/summary SHA-256 is `69b0100aae27...b8e7`/`74827af0d9e6...e164`/ `2b739a495fc8...ec53` report-content/manifest-content SHA-256 is `84e7b3521a54...85c3`/`018af80f1942...154d`. The complete package is `outputs/af_fno/C/anomaly_direct_s0_reflected_spectral_gate_control_v1/`.

Frozen Arm R reduced-channel causal control, 29 July 2026. Contract `config/model_c_reduced_channel_control_v1.json`, SHA-256 `a1b8ce50d1b...5a6c5`, directly tests the highest-ranked unresolved hypothesis. It replaces the 46-channel autoregressive state by ten recovered Bire-facing outputs: surface/mid U, V, and temperature; surface/mid/ bottom `PHIHYD`; and barotropic streamfunction. `PHIHYD` and streamfunction are derived from each full truth snapshot only while creating targets. Thereafter they are predicted directly; no omitted full-state channel is reconstructed or fed back.

All other selected-Model-C choices are fixed: pooled split-1 pointwise anomalies with the 5th-percentile scale floor, direct ten-day future state, width 128, four FNO blocks, 24×16 modes, 10% padding, lift/ projection ratio two, Channel MLP $4C$, the seed-20260724 optimizer and 15,360-step schedule, seven checkpoint times, and the three-step state/increment/rollout/spectral/western-boundary objective with unchanged weights. The unavoidable loss adaptation replaces the absent U/V/temperature/SSH group partition by equal U/V/temperature/`PHIHYD`/ streamfunction groups over the declared ten outputs; this is recorded rather than mislabeled as an exactly identical 46-channel loss.

The transactional 2.2-GB derived cache has shape $3 \times 7200 \times 10 \times 62 \times 62$, logical-state SHA-256 `b24728abf50f...c5aa`, and is bitwise equal to fresh reconstruction at four boundary/split-spanning snapshots. Its consolidated metadata and quality-report SHA-256 values are `3ec92a0b30d6...253`/`b8842dea6f50...a50`. Training-only selection retains the exact 540-rollout, day-10–90 gate and bitwise nine-call reload check. Only after that selection does the same 15-member S0 ensemble, day-2,000 truth, persistence/climatology reductions, and six Figure 3–8 filenames as job 304736 open. Figure 6 is now the direct 46-channel-source versus ten-channel-Arm-R ACC comparison. The descriptive held evaluation runs even if the training gate fails, but a failed training gate forbids promotion.

Six focused tests, lint, shell validation, source and artifact locks, both static preflights, exact normalizer/record hashes, and a finite zero-land native-grid three-call loss smoke test pass. The model has 14,823,642 parameters. Planned lightweight outputs are isolated under `outputs/af_fno/C/anomaly_direct_reduced_channel_bire_s0_inference_v1/`. The complete adjoint campaign remains closed until both the deterministic 10–90-day gate and the bounded, statistically plausible day-2,000 gate are interpreted from the frozen result; post-decorrelation trajectory RMSE need not beat persistence or climatology.

Temporal-interface decision. The active Model C is one-input/one-output in *time*, despite its 51 input and 46 output *channel* counts. Each call receives one 46-channel state together with wind stress, longitude, latitude, mask, and wall distance, then predicts one 46-channel ten-day residual. Three-step autoregressive training exposes the model to its own predictions but does not provide two past states to one call or produce two future states directly.

Samudra instead used consecutive ocean states to predict two later states, arguing that the pair supplies tendency-like context; Bire identified this two-time-step input/output design as a future forecast-horizon comparison rather than using it in the reported FNO. It remains a bounded secondary arm rather than the next automatic job: the streamfunction phase is already accurate through day 90, job 292064 selected the forecast-age branch, and completed job 292293 improved primary AUC without passing the long gate. A later parameter/compute-accounted two-input/two-output temporal ablation must retain the same data, Fourier modes, spatial trunk, conservation projections, bias penalty, and prospective gate. The latter would naturally use 97 external inputs (two 46-channel states plus five forcing/static fields) and 92 dynamic outputs, and would make the eventual Model D/adjoint state carry two consecutive ocean states. Its extra temporal context may reduce recursive phase error, but its larger interface and paired-output consistency are costs, so improvement cannot be assumed from Samudra's different global-OM4 ConvNeXt-UNet setting.

Bire-style presentation characterization. Bire Figure 3 shows barotropic streamfunction truth, prediction, and truth-minus-prediction at days 0–40, followed by additional coarse-resolution rows; Figure 4 shows mean RMSE with 10th–90th percentile shading over 15 randomly chosen members. Its persistence repeats each initial condition at all later leads. Its climatology is described as the pointwise time mean of the MITgcm simulation, but the article does not explicitly state whether evaluation times are excluded from that mean.

Contract `config/model_c_bire_figures_v1.json`, SHA-256 `791c84f12c2d...eb696b`, is frozen before any 100–200-day validation metric. It reproduces those definitions without opening inference: one hash-pinned seed-20260723 checkpoint, 15 prospectively sampled S2 fresh-validation starts, and the first prospectively drawn member for the spatial panels. Figure 3 retains only the native 1° truth/prediction/difference rows. Figure 4 retains only $\Delta t = 10$ days and fixes 0–200 days, red/black/blue prediction/climatology/persistence curves, member mean and 10th–90th percentiles, and requested limits 0–1.2×10^{-1} $\mathrm{m\,s^{-1}}$ for speed, 0–3×10^{-1} °C for SST, and 0–1 $\mathrm{m^2\,s^{-2}}$ for P/ρ. The climatology is the regime-specific pointwise mean over all 5,040 split-1 training snapshots, with nonlinear derived fields averaged after derivation. This is a leakage-free implementation of the paper's formula. The output is descriptive evidence for the already rejected model and cannot select a checkpoint, alter a loss/architecture, or authorize inference. GPU job 291134 completed normally in 51 seconds. For the prospectively selected S2 member, streamfunction RMSE at days 10/20/30/40 is 0.0316/0.0934/0.1580/0.1765 Sv against a truth RMS near 13.34 Sv. Thus the short-horizon circulation fields are very close; the shared paper-style streamfunction scale makes the difference row intentionally faint.

All 15 predictions remain finite through day 200, but the long-horizon curves expose severe recursive drift. At day 200, Model-C/persistence RMSE is 0.07344/0.00480 $\mathrm{m\,s^{-1}}$ for surface speed, 5.981/0.0505 °C for SST, and 3.747/0.0253 $\mathrm{m^2\,s^{-2}}$ for surface pressure. Model mean SST first exceeds its requested 0.3-degree axis at day 100, and surface pressure first exceeds its requested 1 $\mathrm{m^2\,s^{-2}}$ axis at day 160; speed remains inside its requested axis. Preserve this clipping because it is part of the requested paper-matched presentation, and use the numerical archive for the off-scale values. Report/arrays SHA-256 values are `b704b921444d...e300ac5/a73ac9bf2cc9...20b1`; the two PNGs, lightweight summary, and manifest-content SHA-256 `258be6fabe13...827f` are preserved under `outputs/af_fno/C/bire_figures_v1/`. Inference remains sealed.

Because the requested fixed axes visually compress the baselines, preserve that paper-matched figure and add, rather than replace it with, the hash-pinned companion `outputs/af_fno/C/bire_baseline_zoom_v1/`. It reads only the immutable Figure-4 arrays and gives full-range plus baseline-scale SST and surface-P/ρ panels and a complete CSV. SST first exceeds both persistence and climatology at day 20.

Surface P/ρ first exceeds persistence at day 30 and climatology at day 70. Contract/manifest-content SHA-256 values are `2a2324090b83...b2a4a/01433936e406...99063`; the figure/CSV SHA-256 values are `1f1ea2c6f76a...6643f/542e74c02b59...068ce`. This addendum evaluates no model, changes no result, and opens no data.

To expose the visually small streamfunction errors without changing the completed package, freeze the separate contract `config/model_c_bire_streamfunction_leads_v1.json` before evaluating the additional spatial maps. Its SHA-256 is `ccb965e58bdc...ae1934c`. It fixes exactly eight new native-1° figures at days 20, 30, . . . , 90 for the same S2 start 6335 and hash-pinned checkpoint. Each image contains MITgcm truth, Model C, and truth-minus-prediction. Truth and prediction use one common symmetric scale across all leads; each difference uses its own labeled symmetric maximum-error scale and prints RMSE and maximum error. This makes spatial structure visible, but colors in different error panels must be interpreted through their printed scales rather than compared directly. Day 10 remains represented in the original Figure-3 analogue. The new immutable roots are `/bigscratch/mjalabert314/bire_james25_repro/af_fno/models/C/bire_streamfunction_leads_v1/` and `outputs/af_fno/C/bire_streamfunction_leads_v1/`. This is descriptive use of the already opened S2 validation member; inference and every later archive remain sealed.

GPU job 291135 completed this contract normally in 35 seconds. Streamfunction RMSE increases from 0.0934 Sv, or 0.700% of truth RMS, at day 20 to 0.2951 Sv, or 2.211%, at day 90; maximum pointwise error increases from 0.7240 to 1.1713 Sv. Thus the large-scale barotropic circulation remains visually close through day 90 even though the independently measured SST and surface-pressure trajectories drift badly. This variable-specific contrast strengthens the objective/checkpoint-mismatch diagnosis: the entire spatial dynamics have not collapsed uniformly. Preserve report/arrays SHA-256 values `71c6e1dc1f9a...89c206/b4126494d9c0...3bcd48` and all eight hash-bound PNGs under the project output root; manifest-content SHA-256 is `3b74d759d551...450ac7`. The repeated days 20–40 agree with the original package to FP32 roundoff, and no inference data were opened.

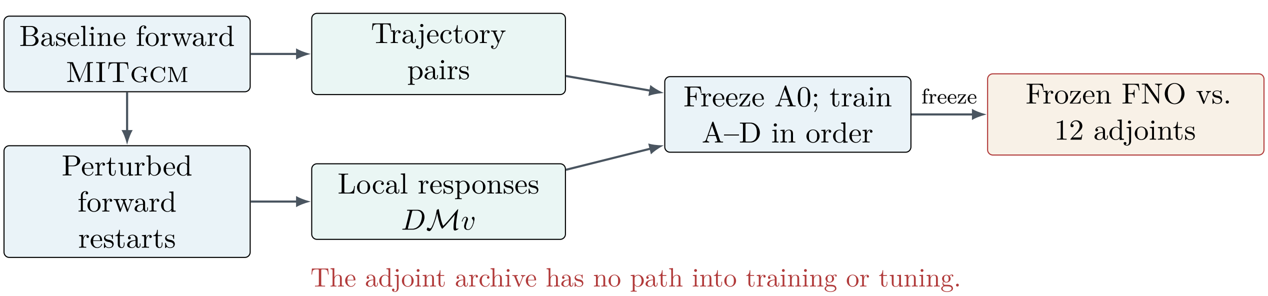

Figure 1: The common trajectory pipeline feeds A0 and A–D; response data enter only D, and the adjoint evaluation remains sealed.

8 Ground-truth MITgcm adjoint suite

8.1 Minimum: twelve integrations

One adjoint integration returns sensitivities to the full initial state and active controls for one scalar objective. It is therefore more efficient to vary objective, horizon, and state than to run an adjoint for every perturbation direction.

Select dates and masks before opening any adjoint result; place dates at least 90 days from split boundaries and one another. If mandatory tests succeed, an optional six-run extension adds 90-day horizons. Before accepting a reference adjoint, run a Taylor remainder or central directional finite-difference check at one

Table 9: Blind matrix: 3 objectives × 2 horizons × 2 initial states = 12 adjoint integrations.

Smooth scalar objective	**Reason**	**Horizons**	**States**
Gaussian interior sea-surface-height average	Broad low-wavenumber response away from walls	10, 30 days	Two unseen dates
Smooth western-boundary sea-surface-height average	Energetic region where spectral/boundary errors matter	10, 30 days	Same dates
Area-weighted barotropic gyre-strength functional	Dynamical circulation measure; avoids nondifferentiable maxima	10, 30 days	Same dates

test state per objective family. This verifies MITgcm but may not alter the emulator.

8.2 Metrics and an operational test

For gradients g_M and g_F, report

$$\text{cosine} = \frac{g_F^T g_M}{\|g_F\|_2 \|g_M\|_2}, \quad \text{angle} = \cos^{-1}(\text{cosine}), \quad \text{relative error} = \frac{\|g_F - g_M\|_2}{\|g_M\|_2}.$$

Convert both gradients to the same physical units and grid with explicit area/volume weights. Report full-state scores and projections onto the directions in Table 6. Direction is primary: optimization can tolerate imperfect magnitude but not a gradient pointing uphill. Also show signed maps, vertical profiles, wavenumber spectra, and sign agreement in the largest-sensitivity region.

Finally, use the emulator gradient to propose a small control update, choose the step from a predeclared short line search, and evaluate the objective with one forward MITgcm run. This tests usefulness without turning adjoints into labels.

9 Evaluation and predeclared success criteria

9.1 Bire et al. as a qualitative forward benchmark

The numerical errors in Bire et al. cannot be transferred to this experiment: their 0.25°, more turbulent double gyre has 248 × 248 points and different state variables, while the present tutorial has 62 × 62 points. We instead predeclare the following *trends* from their complete forward study as the benchmark [1]:

1. Their ten-day map gave the best overall compromise among 5-, 10-, and 30-day maps. It beat persistence and pointwise time-mean climatology at useful short leads; velocity lost deterministic skill first, near roughly three months, while pressure and surface temperature retained useful skill longer.
2. Spatial error grew first in the energetic western-boundary region. At day 60 the ten-day map retained the most credible large-scale and boundary structure; after deterministic phase information was lost, stable long integrations still retained the characteristic anomaly length scales.
3. The learned integrations remained bounded for thousands of model days. The five-day map developed excessive low-wind mesoscale variability, whereas the 30-day map aliased or missed propagating variability whose period was close to the map interval. This makes spectra and longitude–time diagrams necessary alongside scalar RMSE.

4. Long-run maximum streamfunction increased approximately linearly with wind-stress amplitude, close to the expected Sverdrup trend for the ten-day map. The low- and high-intermediate wind cases were held out of training, so their ordering was a real interpolation test rather than a fit diagnostic.
5. Their emulator did *not* reliably steer from an initial condition from one regime after an abrupt change in imposed wind. Initial-condition dynamics dominated the new forcing. We record this negative result explicitly: matching the equilibrium wind ordering is necessary, but a response/adjoint model intended for control should also improve the forcing-switch response.

"Reproduce or improve Bire's trends" below means recover this evidence pattern, not match their plotted values or claim an exact reproduction. A model may decorrelate—a chaotic forecast should—but it must beat both simple baselines before decorrelation, remain bounded, preserve resolved spatial scales, and express the correct forcing ordering. The complete adjoint campaign does not open until at least one modern forward model passes the gate below.

Their hyperparameters are not transferable either

The same caution applies to Bire's optimizer settings, and it has now been demonstrated rather than assumed. Their Table 2 fixes batch size 8, learning rate 0.01, $L = 3$, $C = 128$, $C_{\text{lift}} = C_{\text{proj}} = 256$, $C_{\text{hidden}} = 4C$, GELU, and $\Delta t \in \{5, 10, 30\}$ days — but for a 248×248 grid with $(N_h, N_w) = (64, 64)$ retained modes and $C_{\text{in}}, C_{\text{out}} = 11, 10$. Our configuration keeps their latent proportions but is a very different optimization problem: 62×62 points, $(24, 16)$ modes, and a 46-channel closed state, so the spectral operator carries 384 modes against their 4,096 while the state carries four to five times as many channels. Adopting 0.01 unchanged drove our map to the trivial zero-anomaly solution without ever producing a non-finite loss. Transfer the architecture proportions and the training *protocol* from Bire; re-derive the learning rate for this configuration and declare it separately.

A collapsed model must not be scored as a stable one

That failure also exposed a gap in the criteria below. A map that emits the climatological field is perfectly bounded, has near-unit spectral and integrated-energy ratios, and a per-call gain of one; every long-horizon stability diagnostic in this plan scores it as the best result available. Any claim of improved stability must therefore be accompanied by evidence that the model retains skill: one-step error compared against the zero-anomaly baseline, which equals 1.0 by construction in normalized units, and anomaly correlation at the evaluated leads. Stability statistics alone are not sufficient evidence of a better emulator.

Their split protocol, unlike their hyperparameters, does transfer

Bire's Section 3.2 arrangement was replicated exactly — 6,000 training, 1,200 validation, 1,000 inference timesteps from a 7,200-day record, inference nested as the final 1,000 days of validation, no buffers — and, unlike their learning rate, it carries over to this configuration without modification. Under it the pooled loss-recovery model beats persistence on all three primary fields at short lead (0.435/0.378/0.296) and holds surface pressure below climatology at every lead out to day 2,000, with day-200 anomaly correlation $+0.66/+0.74/+0.71/+0.97$. That is the evidence pattern predeclared above: skill against both baselines at useful short leads, velocity losing deterministic skill first, pressure persisting longest, and a bounded non-diverging tail.

The pattern holds only at short lead, and the criteria above should be read as requiring that. The model exceeds persistence from roughly 170 days and persistence is better at every lead from 700 to 2,000 days on all three fields. Surface pressure stays under climatology throughout, but the pressure climatology degrades by a factor 1.73 across the evaluation window while speed and SST degrade by 1.12 and 1.22, so a ×climatology score at long lead partly measures the baseline rather than the model. Long-horizon claims should quote persistence as well as climatology.

Two cautions attach to it. The inference block is nested inside validation, so it is a replication of the paper's setting rather than a demonstration of prospective skill on disjoint data; the same model under a strictly chronological split with a larger training-to-evaluation gap scores 0.686/0.659/0.623. And the arrangement should be reported alongside any skill number, because with everything else held fixed it moves day-200 error from 2.76/2.27/0.73 to 1.08/0.86/0.24 times climatology.

9.2 Frozen forward evaluation package

Evaluate A0 and Models A–D on identical chronological records and identical 15-member ensembles in each forcing regime. A0 is frozen before I1/I2 are opened and no model is tuned after viewing response or adjoint tests. Persistence and a regime-specific, training-only pointwise mean are the two primary baselines. The current configuration has no imposed seasonal cycle, so calling the latter a daily climatology would be misleading; a day-of-year climatology will be added only if seasonal forcing is added.

Every frozen model must produce the following versioned package. Evaluation contract version 2 adds the deterministic pressure postprocessor; version-1 packages remain historical pre-pressure evidence and are never overwritten. Machine-readable JSON stores definitions, scalar summaries, start indices, hashes, and gate outcomes; compressed NPZ stores the plotted arrays so a figure can be regenerated without rerunning the network.

1. **Deterministic skill:** physical-unit RMSE and pointwise-climatology ACC from 10 to 360 days for U, V, temperature, SSH, surface speed, SST, `PHIHYD` at levels 0, 7, and 14, and the U-derived barotropic transport streamfunction. Curves show ensemble mean and 10–90% range for the model, persistence, and climatology. Report the first lead at which each method loses ACC or baseline skill; never infer stability from the ten-day score alone.
2. **Spatial failures:** common-scale truth/forecast/error snapshots and spatial RMSE maps at 10, 60, 180, and 360 days for all seven Bire-facing diagnostics: surface speed, SST, SSH, the three pressure levels, and the U-derived barotropic transport streamfunction. Report the first four wet cells from the western wall separately, plus full-domain vertical RMSE profiles, to expose boundary and deep-ocean drift.
3. **Stability and scale retention:** one-year trajectories of a surface kinetic-energy proxy, temperature, SSH, and maximum streamfunction; finite-value and land-mask checks; normalized mean/variance drift by depth; radially binned spectra for all seven Bire-facing diagnostics at 10, 90, 180, and 360 days; and a fixed-control-member SSH longitude–time diagram. Spectra are interpreted only over bins with non-negligible truth energy.
4. **Forcing physics:** low/control/high curves of maximum streamfunction, surface speed, SST, and SSH, with model and MITgcm slopes computed on the same ensemble. Once generated, I1/I2 are added as inference-only interpolation points. A separate forcing-switch experiment initializes every branch from the same control states and changes only the prescribed wind; it is clearly labeled as a response diagnostic until matching switched-wind MITgcm trajectories exist.
5. **Audit trail:** checkpoint and dataset hashes, software/device metadata, exact ensemble starts, masks, normalizers, units, climatology definition, spectra, snapshots, maps, and all per-member values needed to reproduce uncertainty bands.

The project separates two scientific claims while retaining the stricter combined authorization rule. The *adjoint-readiness* forward gate is matched to the mandatory 10- and 30-day adjoint horizons. Job 292064 supplies its training-only predictability evidence: damped persistence has SST/SSH RMSE-AUC ratios 0.771/0.954 to plain persistence, SST spatial e-folding times are 23–37 days, and SSH spatial persistence is 248 days in S0 and beyond 720 days in S1/S2. The versioned short gate will therefore include damped persistence and anomaly-scale drift. Job 292293 supplies the original training-only parent at days 10/30, and job 303447 now supplies a substantially stronger anomaly-direct candidate whose fixed S2 primary fields beat both baselines throughout the 10–90-day AUC comparison and remain bounded through day 200. All three independent seeds pass, and the median seed is frozen before inference. Contract `model_c_anomaly_direct_forward_v1` fixes the short gate, its damped-persistence and same-start A0 comparisons, ensemble rule, anomaly-scale bound, and complete long-forward characterization before any held value is read. Passing the versioned 10–30-day determination is retained separately from the 360-day long-rollout claim. Job 303993 passes that short determination and every 10–90-day primary AUC, boundedness/drift, and wind-slope check, but fails the combined contract solely at the mid/bottom-pressure spectral criterion. Under the prospectively frozen rule, the complete response/adjoint campaign opens only when both determinations pass. It therefore remains sealed while the localized training-split spectral attribution and one bounded successor test proceed.

The long-rollout forward gate is passed by a modern model only if all of the following hold:

- over the pre-decorrelation 10–90-day window, it improves on A0 for Bire's three reference-primary indicators—surface speed, SST, and derived surface `PHIHYD`—and has a smaller normalized RMSE area-under-curve than both persistence and training climatology. Its ACC area-under-curve must exceed persistence and the zero climatology reference. The final machine-readable validation contract will freeze the discrete leads and block-bootstrap confidence rule before fresh v2 validation opens; selection is based on these curves rather than a single ten-day crossing. Barotropic streamfunction remains mandatory for circulation/stability and the wind-response gate below. Mid/bottom pressure and every prognostic U/V/temperature level are mandatory reported diagnostics but are not multiplied into independent copies of the same pass criterion;
- learned SSH is reported explicitly with RMSE, ACC, drift, maps, and block-bootstrap uncertainty, but it is not a standalone validation veto. Bire did not predict or score SSH separately, and the derived surface-pressure diagnostic already contains the $g\eta$ barotropic contribution. SSH may still reject a model through instability, gross bias/variance drift, or failure of the common physical package, not merely because its ten-day persistence ratio is infinitesimally above one;
- its 360-day rollout is finite, respects the land mask, has no wet-cell normalized state magnitude above 20, and has no group-mean 360-day drift larger than two training standard deviations relative to the matching MITgcm ensemble;
- at 360 days its resolved-bin median spectral-energy ratio lies in $[0.25, 4]$ for each Bire-facing field; phase agreement is not required after decorrelation;
- the fitted low/control/high maximum-streamfunction slope has the same sign as MITgcm and magnitude within a factor of two. The same requirement is later applied with I1/I2 inserted, without refitting or retuning the emulator;
- the forcing-switch response has the physically expected sign by one year. Until switched-wind MITgcm truth is generated, this last item is reported as provisional and cannot by itself establish a pass.

Failure of a subcriterion is retained and plotted, not averaged away. In particular, a lower aggregate normalized loss cannot compensate for temperature or SSH being worse than persistence, a blow-up, or reversed wind sensitivity. If neither A nor B passes, the project proceeds only through the predeclared Model C forward-selection stage. The long-rollout claim remains closed unless C, or an explicitly versioned successor, passes these criteria. The mandatory 10/30-day perturbation/adjoint campaign follows the separate horizon-matched amendment above; optional 90-day adjoints remain conditional on the long-rollout gate.

The three pressure fields are correlated derived views of temperature and SSH, not three additional independent trials. They are nevertheless required because Bire et al. reported them directly and because a model can have acceptable bulk temperature/SSH error while producing vertically accumulated hydrostatic errors that are unsuitable for pressure-based objectives. Gate failures are therefore interpreted field by field; pressure does not alter training or inflate the learned state dimension.

9.3 Response and adjoint evaluation package

The derivative study uses the predeclared dates, 12 objective–horizon cases, direction families, regional masks, physical inner products, and perturbation amplitudes. All models see the same states and perturbations. Three final seeds are used for the paired A/B/C/D comparison if affordable; seed spread is shown rather than folded into the test count. For each case save both physical-unit and normalized arrays, including state and control perturbations, masks, objective weights, JVPs, gradients, Taylor sequences, and line-search outcomes. Required displays and tables are:

1. **Perturbation validity:** central-difference linearity versus perturbation amplitude, plus/minus symmetry, numerical noise floor, and Taylor remainder slopes, separated by variable, depth, direction family, boundary/interior region, and horizon.
2. **Forward responses:** true, emulator, and difference response maps on common symmetric colour scales; predicted-versus-true directional-response scatter; relative response error, cosine, and gain ratio. Show every case as well as bootstrap intervals for the paired aggregate; do not hide sign failures behind the median.
3. **Adjoint fields:** signed MITgcm, emulator, and difference maps; depth slices and vertical profiles; full physical-space and declared projected-subspace relative error, cosine angle, magnitude ratio, sign agreement, and top-sensitivity overlap. Plot error versus wall distance and depth, and compare resolved gradient spectra/coherence to distinguish phase, amplitude, and boundary errors.
4. **Operator consistency:** verify the duality identity $q^T(Jv) = v^T(J^Tq)$ for the emulator, compare automatic differentiation with finite differences, and report how ten-day Jacobian errors compose at 30 days.
5. **Control usefulness:** for each objective plot the predeclared line-search step, emulator-predicted ΔJ, forward-MITgcm ΔJ, and constraint residual. Store the full objective-versus-step curve, not only the chosen update.

Predeclared derivative targets, not promises, are:

- **Response:** D lowers blind response error by at least 25% versus its forward parent C while ten-day state error rises by no more than 5%.
- **Adjoint:** D improves median projected-gradient cosine by at least 0.2 versus C. Useful targets are median cosine above 0.8 at ten days and 0.6 at 30 days.
- **Usefulness:** at least 80% of predeclared emulator-gradient control steps lower the objective in forward MITgcm.

Failure remains diagnostic. Better one-step JVPs but poor 30-day adjoints implicate Jacobian composition; no JVP improvement implicates perturbation scale or subspace; interior-only success implicates boundary representation; good angles but wrong gains implicate calibration rather than sensitivity direction.

10 Ten-week execution plan

Table 10: Long MITgcm runs begin immediately and continue alongside code development.

Week	Activity	Output	Decision gate
1	Reproduce tutorial; benchmark; launch S0	Versioned setup, restart test, wall-clock/model-year	Runs are reproducible
2	Complete S0–S2; preprocess state maps	Common data store, masks, transpose-checked operators	Training trajectories pass QC
3	Audit and freeze A0; train ten-day baseline	Architecture manifest, small-sample overfit, A0 checkpoint	Adapted Bire map learns signal
4	Train modern Model A; spectrum study	A0/A forward comparison and baselines	Record controlled baseline
5	Train and freeze B	Rollout, spectrum, and boundary-loss ablation	Diagnose remaining forward failures
6	Select C; diagnose rejection; expand data	Learning curves, three seeds, data-adequacy audit, version-2 trajectories	Successor forward gate must pass
7	Run I1/I2; perturbation pilot and response data	Interpolation and linearity plots; training/validation responses	Perturbations resolved and local
8	Train D; freeze derivative comparison	C/D response ablation; sealed checkpoints and scripts	State skill preserved within 5%
9	Generate/verify adjoints; gradient and control tests	Taylor checks, maps, angles, and true objective changes	Open only after all freezes
10	Robustness, report, and handoff	Reproducible repository and manifest	Complete minimum claim

11 Risks and fallbacks

The 0.25° configuration is a stretch goal only after C passes at 1°. A valuable extension is resolution transfer at 0.5° or 0.25°, testing both values and local responses. It must not displace the mandatory adjoint comparison.

12 Deliverables and reproducibility

The internship should leave:

1. version-controlled MITgcm configurations for S0–S2, I1–I2, perturbed restarts, and the 12 adjoint cases;
2. a manifest with checksums, units, grids, masks, split dates, perturbation vectors, and configuration hashes;
3. a frozen A0 architecture manifest and pinned NeuralOperator 2.0 pipelines for A0 and Models A–D;
4. separate sealed archives for forward training data and comparison-only adjoints;
5. scripts for forward, response, adjoint, and control-update metrics;
6. a report stating the validated state, subspace, horizons, objectives, and failures.

Table 11: Safeguards preserve a useful ten-week result.

Risk	**Early diagnostic**	**Fallback**
Spin-up cost or deep drift	Trends after year 50; measured cost	Scope first claim to upper ocean or use a documented equilibrated restart
Bire reconstruction ambiguity	Architecture audit exposes paper/repository conflicts	Freeze and label A0 as adapted; time-box historical repair
46-channel memory pressure	100-sample overfit and GPU profile	Reduce width, then TFNO, then vertical representation
FFT wraps across walls	Seam and boundary-response errors	More padding; Fourier continuation; local convolution and wall distance
Perturbation noise/nonlinearity	Central-difference pilot	Change ε by family before freezing it
Good ten-day, poor 30-day gradient	Horizon-separated metrics	Strengthen three-step training; report validated horizon honestly
Adjoint build delayed	Week-1 compilation smoke test	Deliver complete forward-response result; adjoint remains follow-on
Sampled directions too narrow	Good projected but poor full-map score	Add directions from response residuals as a labeled extension

Record seeds, package lockfile, model checksum, normalization statistics, optimizer state, splits, and timing. Unit tests should cover staggered-grid interpolation and its transpose, gradient denormalization, masking, response construction, and a toy autodiff/adjoint identity.

13 Final scope statement

This project is feasible because it makes both the architecture-transfer and adjoint questions small and testable. It does not rebuild every undocumented detail of Bire et al.; A0 preserves the recoverable architecture and labels every forced adaptation, then the A–D ladder separates controlled modernization, forward selection, and derivative supervision on a documented MITgcm case. The original plan prioritizes 224 short, information-rich perturbations, but the completed forward evidence authorizes 30 additional trajectory years before those response experiments. It then reserves 12 expensive adjoints for a blind comparison.

The contribution is an empirical test of a transferable principle: *local forward responses can constrain a neural operator's derivatives enough to improve adjoints for unseen objectives, without adjoint supervision.* The baseline, ablations, sealed evaluation, and explicit subspace claim make the outcome useful whether the hypothesis succeeds or fails.

References

[1] S. Bire, B. Lütjens, K. Azizzadenesheli, A. Anandkumar, and C. Hill, *Ocean Emulation With Fourier Neural Operators: Double Gyre*, *Journal of Advances in Modeling Earth Systems*, 17, e2023MS004137, 2025. doi:10.1029/2023MS004137.

[2] V. Duruisseaux, J. Kossaifi, and A. Anandkumar, *Fourier Neural Operators Explained: A Practical Perspective*, arXiv:2512.01421v2, 2026. arXiv:2512.01421.

[3] MITgcm contributors, *Baroclinic Ocean Gyre*, MITgcm documentation. mitgcm.readthedocs.io: baroclinic gyre.

[4] S. Bire and contributors, *oceanfourcast*, public source repository. github.com/suyashbire1/oceanfourcast.

[5] NeuralOperator contributors, *NeuralOperator: Learning in infinite dimensions*, development documentation and repository. neuraloperator.github.io/dev; github.com/neuraloperator/neuraloperator.

[6] NASA AMT Tiger Team, *IMVIGORATE: Guided Operational Rigor for Assimilative Trusted Emulation*, internal project proposal, 2026.